

Scalar Quantity:

All those quantities which requires only magnitude for complete description are called **scalar quantities** e.g. time, density, temperature and length etc.

Vector Quantity:

All those quantities which requires magnitude as well as direction for their complete description are called **vector quantities** e.g. weight, force, momentum, displacement, velocity etc.

Geometric Interpretation of Vector:

Geometrically, a vector is represented by a directed line segment $\overrightarrow{AB}$ with A its initial point and B its terminal point.

Magnitude of a Vector:

The magnitude or length or norm of vector $\overrightarrow{AB}$ is its absolute value and is written as $|\overrightarrow{AB}|$.

Unit Vector:

A unit vector $\hat{v}$ (read as v hat) of a given vector v is a vector with magnitude one and direction same as vector v . Mathematically

$$\text{Unit Vector} = \frac{\text{Vector}}{\text{Magnitude of Vector}}$$

$$\text{i.e. } \hat{v} = \frac{v}{|v|}$$

Null Vector:

A vector whose terminal point coincides with its initial point is called **null** or **zero vector**.

Negative of a Vector:

Two vectors u and v are called negative of each other, if they have same magnitude but opposite direction.

Multiplication of a Vector by a scalar (Number):

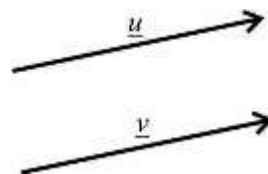
Multiplication of a vector v by a scalar ' n ' is a vector whose magnitude is n times that of ' v ' i.e. nv .

(i) If n is positive, then v and nv are in the same direction.

(ii) If n is negative, then v and nv are in opposite directions.

Equal Vectors:

Two vectors u and v are called equal vectors, if they have same magnitude and same direction.



Parallel Vectors:

Two vectors $\underline{u}$ and $\underline{v}$ are parallel if and only if they are non-zero scalar multiple of each other i.e.

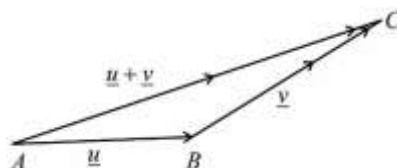
$$\underline{u} = \lambda \underline{v}, \lambda \neq 0$$

Triangle Law of Addition:

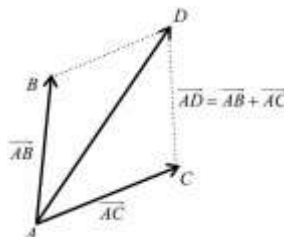
If two vectors $\underline{u}$ and $\underline{v}$ are represented by the two sides $\overrightarrow{AB}$ and $\overrightarrow{BC}$ of a triangle such that the terminal point of $\underline{u}$ coincide with the initial point of $\underline{v}$, then the third side $\overrightarrow{AC}$ of the triangle gives vector sum $\underline{u} + \underline{v}$, that is

$$\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$$

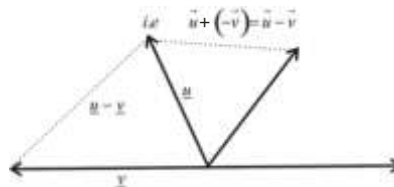
$$\Rightarrow \underline{u} + \underline{v} = \overrightarrow{AC}$$

**Parallelogram Law of Addition:**

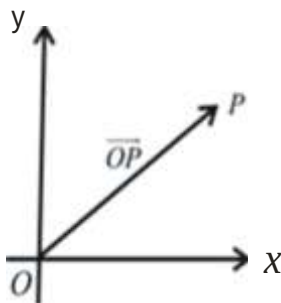
If two vector $\underline{u}$ and $\underline{v}$ are represented by two adjacent sides $\overrightarrow{AB}$ and $\overrightarrow{AC}$ of a parallelogram as shown in the figure, then diagonal $\overrightarrow{AD}$ give the sum or resultant of $\overrightarrow{AB}$ and $\overrightarrow{AC}$, that is $\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{AC} = \underline{u} + \underline{v}$

**Subtraction of two vectors:**

Let $\underline{u}$ and $\underline{v}$ are non-zero vector then subtraction of $\underline{v}$ and $\underline{u}$ is defined as the addition of $\underline{u}$ and $-\underline{v}$ i.e. $\underline{u} + (-\underline{v}) = \underline{u} - \underline{v}$

**Position Vector:**

A vector which describes the location of a point w.r.t origin is called position vector. The vector, whose initial point is the origin O and whose terminal point P is called the position vector of the point P and is written as $\overrightarrow{OP}$

**Vector in a Plane:**

Let R be a set of real numbers. The Cartesian plane is defined to be the

$$R^2 = \{(x, y) : x, y \in R\}.$$

The Unit Vectors $\underline{i}$, $\underline{j}$:

$\underline{i}$ and $\underline{j}$ are called unit vectors along x-axis and y-axis respectively.

They are written as

$$\underline{i} = [1, 0], \underline{j} = [0, 1]$$

$$|\underline{i}| = \sqrt{(1)^2 + (0)^2} = 1$$

$$|\underline{j}| = \sqrt{(0)^2 + (1)^2} = 1$$

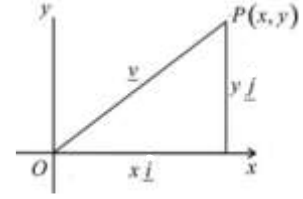
A vector $\vec{v}$ can be written as

$$\underline{v} = [x, y] = [x, 0] + [0, y] = x[1, 0] + y[0, 1] = x\underline{i} + y\underline{j}$$

Similarly, sum of two vectors $\underline{u}$ and $\underline{v}$ can be written as

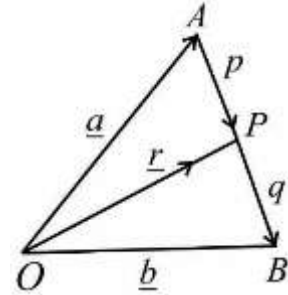
$$\underline{u} = [x_1, y_1], \underline{v} = [x_2, y_2]$$

$$\underline{u} + \underline{v} = [x_1 + x_2, y_1 + y_2] = (x_1 + x_2)\underline{i} + (y_1 + y_2)\underline{j}$$

**The Ratio Formula:**

Let A and B are two points whose position vectors are $\underline{a}$ and $\underline{b}$ respectively. If a point P divides AB in the ratio $p:q$, then the position vector of P is given by

$$\underline{r} = \frac{q\underline{a} + p\underline{b}}{q + p}$$

**Proof:**

Given $\underline{a}$ and $\underline{b}$ are position vectors of the points A and B respectively. Let $\underline{r}$ be the position vector of the point P

which divides the line segment $\overline{AB}$ in the ratio $p:q$.

That is

$$m\overline{AP} : m\overline{PB} = p : q$$

so

$$\frac{m\overline{AP}}{m\overline{PB}} = \frac{p}{q}$$

$$\Rightarrow q(m\overline{AP}) = p(m\overline{PB})$$

$$\text{Thus } q(\overline{AP}) = p(\overline{PB})$$

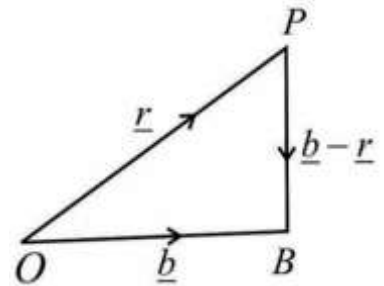
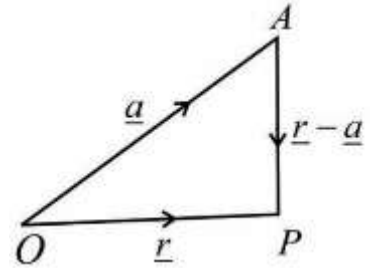
$$\Rightarrow q(\underline{r} - \underline{a}) = p(\underline{b} - \underline{r})$$

$$\Rightarrow q\underline{r} - q\underline{a} = p\underline{b} - p\underline{r}$$

$$\Rightarrow p\underline{r} + q\underline{r} = p\underline{b} + q\underline{a}$$

$$\Rightarrow \underline{r}(p + q) = q\underline{a} + p\underline{b}$$

$$\Rightarrow \underline{r} = \frac{q\underline{a} + p\underline{b}}{q + p}$$



Corollary:

If P is the midpoint of AB , then $p:q=1:1$

$$\therefore \text{Position vector of } P = \underline{r} = \frac{\underline{a} + \underline{b}}{2}$$

Example:

Use vectors, to prove that the diagonals of a parallelogram bisect each other.

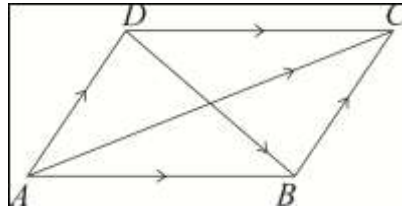
Solution:

Let $A(\underline{a})$, $B(\underline{b})$, $C(\underline{c})$ and $D(\underline{d})$ be the vertices of parallelogram $ABCD$

$$\therefore \overrightarrow{AB} = \overrightarrow{DC}$$

$$\underline{b} - \underline{a} = \underline{c} - \underline{d}$$

$$\text{Or } \underline{b} + \underline{d} = \underline{a} + \underline{c} \dots (i)$$



Let M be the mid-point of diagonal $\overrightarrow{AC}$

$$\text{Then } M = \frac{\underline{a} + \underline{c}}{2}$$

And N be the mid-point of diagonal $\overrightarrow{BD}$

$$N = \frac{\underline{b} + \underline{d}}{2}$$

If the diagonals of parallelogram bisect each other.

Then, mid-point of diagonal $\overrightarrow{AC}$ = mid-point of diagonal $\overrightarrow{BD}$

From equation (i) $\underline{b} + \underline{d} = \underline{a} + \underline{c}$

$$\begin{aligned} \Rightarrow \frac{\underline{b} + \underline{d}}{2} &= \frac{\underline{a} + \underline{c}}{2} \\ \Rightarrow N &= M \end{aligned}$$

i.e. diagonals of parallelogram bisect each other.

EXERCISE 7.1

Q.1 Write the vector $\overrightarrow{PQ}$ in the form $x\mathbf{i} + y\mathbf{j}$.

(i) $P = (2, 3)$, $Q(6, -2)$

Solution:

$$\begin{aligned} P &= (2, 3), \quad Q(6, -2) \\ \overrightarrow{PQ} &= \overrightarrow{OQ} - \overrightarrow{OP} \quad (\text{Where } O \text{ is origin}) \\ &= (6\mathbf{i} - 2\mathbf{j}) - (2\mathbf{i} + 3\mathbf{j}) \\ &= (6-2)\mathbf{i} + (-2-3)\mathbf{j} \end{aligned}$$

$$\boxed{\overrightarrow{PQ} = 4\mathbf{i} - 5\mathbf{j}}$$

(ii) $P = (0, 5)$, $Q(-1, -6)$

$$\begin{aligned} \overrightarrow{PQ} &= \overrightarrow{OQ} - \overrightarrow{OP} \\ &= (-\mathbf{i} - 6\mathbf{j}) - (0\mathbf{i} + 5\mathbf{j}) \\ &= (-1-0)\mathbf{i} + (-6-5)\mathbf{j} \end{aligned}$$

$$\boxed{\overrightarrow{PQ} = -\mathbf{i} - 11\mathbf{j}}$$

Q.2 Find the magnitude of the vector $\mathbf{u}$:

(i) $\mathbf{u} = 2\mathbf{i} - 7\mathbf{j}$

Solution:

$$\begin{aligned} \mathbf{u} &= 2\mathbf{i} - 7\mathbf{j} \\ |\mathbf{u}| &= |2\mathbf{i} - 7\mathbf{j}| \\ &= \sqrt{2^2 + (-7)^2} \\ &= \sqrt{4 + 49} \end{aligned}$$

$$\boxed{|\mathbf{u}| = \sqrt{53}}$$

(ii) $\mathbf{u} = \mathbf{i} + \mathbf{j}$

Solution:

$$\begin{aligned} |\mathbf{u}| &= |\mathbf{i} + \mathbf{j}| \\ &= \sqrt{(1)^2 + (1)^2} \end{aligned}$$

$$\boxed{|\mathbf{u}| = \sqrt{2}}$$

(iii) $\mathbf{u} = 3\mathbf{i} - 4\mathbf{j}$

Solution:

$$\begin{aligned} |\mathbf{u}| &= |3\mathbf{i} - 4\mathbf{j}| \\ &= \sqrt{(3)^2 + (-4)^2} \end{aligned}$$

$$= \sqrt{9+16}$$

$$= \sqrt{25}$$

$$\boxed{|\mathbf{u}| = 5}$$

Q.3 If $\mathbf{u} = 2\mathbf{i} - 7\mathbf{j}$, $\mathbf{v} = \mathbf{i} - 6\mathbf{j}$ and $\mathbf{w} = -\mathbf{i} + \mathbf{j}$. Find the following vectors.

(i) $\mathbf{u} + \mathbf{v} - \mathbf{w}$

Solution:

$$\begin{aligned} \mathbf{u} + \mathbf{v} - \mathbf{w} &= (2\mathbf{i} - 7\mathbf{j}) + (\mathbf{i} - 6\mathbf{j}) - (-\mathbf{i} + \mathbf{j}) \\ &= 4\mathbf{i} - 14\mathbf{j} \end{aligned}$$

(ii) $2\mathbf{u} - 3\mathbf{v} + 4\mathbf{w}$

Solution:

$$\begin{aligned} 2\mathbf{u} - 3\mathbf{v} + 4\mathbf{w} &= 2(2\mathbf{i} - 7\mathbf{j}) - 3(\mathbf{i} - 6\mathbf{j}) + 4(-\mathbf{i} + \mathbf{j}) \\ &= 4\mathbf{i} - 14\mathbf{j} - 3\mathbf{i} + 18\mathbf{j} - 4\mathbf{i} + 4\mathbf{j} \\ &= -3\mathbf{i} + 8\mathbf{j} \end{aligned}$$

(iii) $\frac{1}{2}\mathbf{u} + \frac{1}{2}\mathbf{v} + \frac{1}{2}\mathbf{w}$

Solution:

$$\begin{aligned} \frac{1}{2}\mathbf{u} + \frac{1}{2}\mathbf{v} + \frac{1}{2}\mathbf{w} &= \frac{1}{2}[\mathbf{u} + \mathbf{v} + \mathbf{w}] \\ &= \frac{1}{2}[(2\mathbf{i} - 7\mathbf{j}) + (\mathbf{i} - 6\mathbf{j}) + (-\mathbf{i} + \mathbf{j})] \\ &= \frac{1}{2}(2\mathbf{i} - 12\mathbf{j}) \\ &= \mathbf{i} - 6\mathbf{j} \end{aligned}$$

Q.4 Find the sum of the vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ given the four points $A(1, -1)$, $B(2, 0)$, $C(-1, 3)$ and $D(-2, 2)$.

Solution:

$$\begin{aligned} \overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} \quad \text{where "O" is origin} \\ &= (2\mathbf{i} + 0\mathbf{j}) - (\mathbf{i} - \mathbf{j}) \\ &= \mathbf{i} + \mathbf{j} \end{aligned}$$

$$\begin{aligned}\overrightarrow{CD} &= \overrightarrow{OD} - \overrightarrow{OC} \\ &= (-2\hat{i} + 2\hat{j}) - (-\hat{i} + 3\hat{j}) \\ &= -\hat{i} - \hat{j}\end{aligned}$$

Now,

$$\begin{aligned}\overrightarrow{AB} + \overrightarrow{CD} &= (\hat{i} + \hat{j}) + (-\hat{i} - \hat{j}) \\ &= 0\hat{i} + 0\hat{j} \text{ Null vector}\end{aligned}$$

Q.5 Find the vector from the point A to the origin where $\overrightarrow{AB} = 4\hat{i} - 2\hat{j}$ and B is the point (-2,5).

Solution:

$$\overrightarrow{OB} = -2\hat{i} + 5\hat{j}$$

We know that

$$\begin{aligned}\overrightarrow{AO} &= \overrightarrow{AB} + \overrightarrow{BO} \\ &= \overrightarrow{AB} - \overrightarrow{OB} \\ &= 4\hat{i} - 2\hat{j} - (-2\hat{i} + 5\hat{j})\end{aligned}$$

$$\boxed{\overrightarrow{AO} = 6\hat{i} - 7\hat{j}}$$

Q.6 Find a unit vector in the direction of the vector given below:

(i) $\underline{v} = 2\hat{i} - \hat{j}$

Solution:

$$\begin{aligned}\underline{v} &= 2\hat{i} - \hat{j} \\ |\underline{v}| &= |2\hat{i} - \hat{j}| \\ &= \sqrt{(2)^2 + (-1)^2} \\ &= \sqrt{4+1} \\ &= \sqrt{5} \\ \hat{\underline{v}} &= \frac{\underline{v}}{|\underline{v}|} \\ &= \frac{2\hat{i} - \hat{j}}{\sqrt{5}}\end{aligned}$$

$$\boxed{\hat{\underline{v}} = \frac{2}{\sqrt{5}}\hat{i} - \frac{1}{\sqrt{5}}\hat{j}}$$

(ii) $\underline{v} = \frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}$

Solution:

$$\begin{aligned}|\underline{v}| &= \left| \frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j} \right| \\ &= \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \\ &= \sqrt{\frac{1}{4} + \frac{3}{4}} \\ &= \sqrt{\frac{4}{4}} = 1\end{aligned}$$

$$\begin{aligned}\hat{\underline{v}} &= \frac{\underline{v}}{|\underline{v}|} \\ &= \frac{\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}}{1}\end{aligned}$$

$$\boxed{\hat{\underline{v}} = \frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}}$$

(iii) $\hat{\underline{v}} = -\frac{\sqrt{3}}{2}\hat{i} - \frac{1}{2}\hat{j}$

Solution:

$$\begin{aligned}|\underline{v}| &= \sqrt{\left(-\frac{\sqrt{3}}{2}\right)^2 + \left(-\frac{1}{2}\right)^2} \\ &= \sqrt{\frac{3}{4} + \frac{1}{4}} = \sqrt{\frac{4}{4}} = 1\end{aligned}$$

$$\hat{\underline{v}} = \frac{\underline{v}}{|\underline{v}|} = \frac{-\frac{\sqrt{3}}{2}\hat{i} - \frac{1}{2}\hat{j}}{1}$$

$$\boxed{\hat{\underline{v}} = -\frac{\sqrt{3}}{2}\hat{i} - \frac{1}{2}\hat{j}}$$

Q.7 If A, B and C are respectively the points $(2, -4), (4, 0)$ and $(1, 6)$. Use vector method to find the coordinates of the point D if:

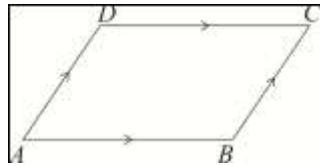
(i) $ABCD$ is parallelogram

Solution:

Let $D(a, b)$ be required vertex

Since $ABCD$ is parallelogram

$$\therefore \overrightarrow{AD} = \overrightarrow{BC}$$



$$(a-2)\underline{i} + (b+4)\underline{j} = (1-4)\underline{i} + (6-0)\underline{j}$$

$$a-2 = -3 \quad b+4 = 6$$

$$a = -3+2 \text{ and } b = 6-4$$

$$a = -1, \quad b = 2$$

$$\text{Thus } \boxed{D = (-1, 2)}$$

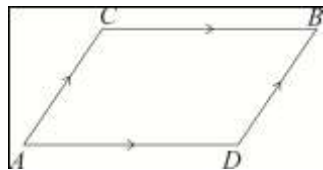
(ii) $ADBC$ is parallelogram

Solution:

Let $D(a, b)$ be required vertex

Since $ADBC$ is parallelogram

$$\therefore \overrightarrow{AD} = \overrightarrow{CB}$$



$$(a-2)\underline{i} + (b+4)\underline{j} = (4\underline{i} + 0\underline{j}) - (\underline{i} + 6\underline{j})$$

$$(a-2)\underline{i} + (b+4)\underline{j} = 3\underline{i} - 6\underline{j}$$

$$a-2 = 3 \text{ and } b+4 = -6$$

$$a = 5 \text{ and } b = -10$$

$$\text{Thus } \boxed{D(5, -10)}$$

Q.8 If B, C and D are respectively $(4, 1), (-2, 3)$ and $(-8, 0)$.

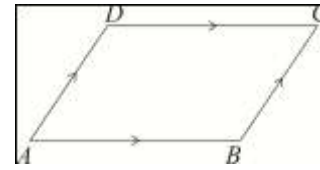
Use vector method to find the coordinates of the point.

(i) A if $ABCD$ is a parallelogram

Solution:

Let $A(a, b)$ be required vertex

Since $ABCD$ is a parallelogram



$$\therefore \overrightarrow{AD} = \overrightarrow{BC}$$

$$(-8-a)\underline{i} + (0-b)\underline{j} = (-2\underline{i} + 3\underline{j}) - (4\underline{i} + 1\underline{j})$$

$$(-8-a)\underline{i} + (0-b)\underline{j} = (-2-4)\underline{i} + (3-1)\underline{j}$$

$$(-8-a)\underline{i} - b\underline{j} = -6\underline{i} + 2\underline{j}$$

$$-8-a = -6, \quad -b = 2$$

$$-8+6 = a, \quad b = -2$$

$$-2 = a$$

$$\text{Thus } \boxed{A(-2, -2)}$$

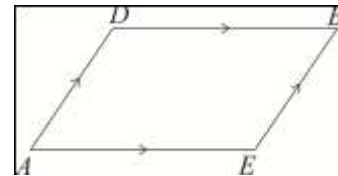
(ii) E if $AEBD$ is a parallelogram

Solution:

Let $E(a, b)$ be required vertex

Since $AEBD$ is parallelogram

$$\therefore \overrightarrow{AE} = \overrightarrow{DB}$$



$$(a\underline{i} + b\underline{j}) - (-2\underline{i} - 2\underline{j}) = (4\underline{i} + \underline{j}) - (-8\underline{i} + 0\underline{j})$$

$$(a+2)\underline{i} + (b+2)\underline{j} = (4+8)\underline{i} + (1-0)\underline{j}$$

$$(a+2)\underline{i} + (b+2)\underline{j} = 12\underline{i} + \underline{j}$$

$$a+2 = 12 \text{ and } b+2 = 1$$

$$a = 10, \quad b = -1$$

$$\text{Thus } \boxed{E(10, -1)}$$

Q.9 If O is the origin and $\overrightarrow{OP} = \overrightarrow{AB}$, find the point P when A and B are $(-3, 7)$ and $(1, 0)$ respectively.

Solution:

Let $P(a, b)$ be required vertex

$$\text{As } \overrightarrow{OP} = \overrightarrow{AB}$$

$$a\underline{i} + b\underline{j} = (\underline{i} + 0\underline{j}) - (-3\underline{i} + 7\underline{j})$$

$$a\mathbf{i} + b\mathbf{j} = (1+3)\mathbf{i} + (0-7)\mathbf{j}$$

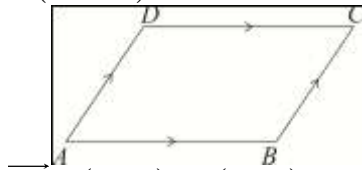
$$a = 4, \quad b = -7$$

Thus $\boxed{P(4, -7)}$

Q.10 Use vectors to show that $ABCD$ is a parallelogram when the points A, B, C and D are respectively $(0,0), (a,0), (b,c)$ and $(b-a,c)$.

Solution:

$A(0,0), B(a,0), C(b,c)$ and $D(b-a,c)$



$$\overrightarrow{AB} = (a-0)\mathbf{i} + (0-0)\mathbf{j}$$

$$= a\mathbf{i} + 0\mathbf{j}$$

$$\overrightarrow{DC} = (b-b+a)\mathbf{i} + (c-c)\mathbf{j}$$

$$= a\mathbf{i} + 0\mathbf{j}$$

$$\overrightarrow{AD} = (b-a-0)\mathbf{i} + (c-0)\mathbf{j}$$

$$= (b-a)\mathbf{i} + c\mathbf{j}$$

$$\overrightarrow{BC} = (b-a)\mathbf{i} + (c-0)\mathbf{j}$$

$$= (b-a)\mathbf{i} + c\mathbf{j}$$

$$\therefore \overrightarrow{AB} = \overrightarrow{DC} \quad \text{and} \quad \overrightarrow{AD} = \overrightarrow{BC}$$

$$\therefore \overrightarrow{AB} \parallel \overrightarrow{DC} \quad \text{and} \quad \overrightarrow{AD} \parallel \overrightarrow{BC}$$

Hence $ABCD$ is a parallelogram.

Q.11 If $\overrightarrow{AB} = \overrightarrow{CD}$. Find the coordinates of the point A when points B, C, D are $(1,2), (-2,5), (4,11)$ respectively.

Solution:

Let $A(a,b)$ be required vertex

$$\overrightarrow{AB} = \overrightarrow{CD}$$

$$(1\mathbf{i} + 2\mathbf{j}) - (a\mathbf{i} + b\mathbf{j}) = (4\mathbf{i} + 11\mathbf{j}) - (-2\mathbf{i} + 5\mathbf{j})$$

$$(1-a)\mathbf{i} + (2-b)\mathbf{j} = (4+2)\mathbf{i} + (11-5)\mathbf{j}$$

$$(1-a)\mathbf{i} + (2-b)\mathbf{j} = (6\mathbf{i} + 6\mathbf{j})$$

$$1-a = 6 \quad \text{and} \quad 2-b = 6$$

$$a = -5 \quad \text{and} \quad b = -4$$

Thus $\boxed{A(-5, -4)}$

Q.12 Find the position vectors of the point of division of the line segment joining the following pair of points, in the given ratio:

- (i) Point C with position vector $2\mathbf{i} - 3\mathbf{j}$ and point D with position vector $3\mathbf{i} + 2\mathbf{j}$ in the ratio 4:3

Solution:

Let $\mathbf{r}$ be the position vector of the point P which divides CD in ratio 4:3 i.e. $\overrightarrow{OP} = \mathbf{r}$

$$\mathbf{r} = \frac{qa + pb}{p+q}$$

$$\mathbf{r} = \frac{3(2\mathbf{i} - 3\mathbf{j}) + 4(3\mathbf{i} + 2\mathbf{j})}{3+4}$$

$$\mathbf{r} = \frac{6\mathbf{i} - 9\mathbf{j} + 12\mathbf{i} + 8\mathbf{j}}{7} = \frac{18\mathbf{i} - \mathbf{j}}{7}$$

$$\boxed{\mathbf{r} = \frac{18}{7}\mathbf{i} - \frac{1}{7}\mathbf{j}}$$

- (ii) Point E with position vector $5\mathbf{j}$ and point F with position vector $4\mathbf{i} + \mathbf{j}$ in ratio 2:5.

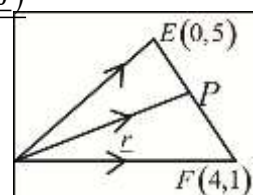
Solution:

Let $\mathbf{r}$ be the position vector of point P which divides E and F into ratio 2:5

$$\mathbf{r} = \frac{5(0\mathbf{i} + 5\mathbf{j}) + 2(4\mathbf{i} + \mathbf{j})}{2+5}$$

$$\mathbf{r} = \frac{(25\mathbf{j} + 8\mathbf{i} + 2\mathbf{j})}{7}$$

$$\boxed{\mathbf{r} = \frac{8}{7}\mathbf{i} + \frac{27}{7}\mathbf{j}}$$



Q.13 Prove that the line segment joining the mid-point of two sides of a triangle is parallel to third side and half as long.

Solution:

Let $A(\underline{a}), B(\underline{b})$ and $C(\underline{c})$ be the vertices of triangle ABC

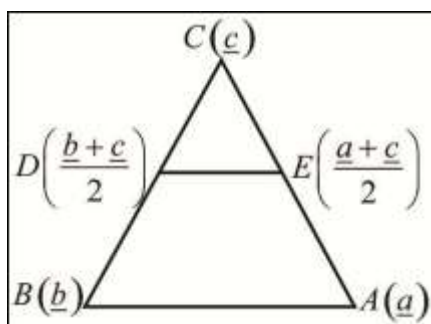
Let D and E are mid points of sides BC and AC respectively.

$$D\left(\frac{\underline{b}+\underline{c}}{2}\right) \text{ and } E\left(\frac{\underline{a}+\underline{c}}{2}\right)$$

$$\text{Now } \overrightarrow{AB} = \underline{b} - \underline{a}$$

$$\overrightarrow{ED} = \left(\frac{\underline{b}+\underline{c}}{2}\right) - \left(\frac{\underline{a}+\underline{c}}{2}\right)$$

$$\begin{aligned} \overrightarrow{ED} &= \frac{\underline{b}+\underline{c}-\underline{a}-\underline{c}}{2} \\ &= \frac{\underline{b}-\underline{a}}{2} \end{aligned}$$



$$\overrightarrow{ED} = \frac{1}{2}(\underline{b} - \underline{a})$$

$$\text{Thus } \overrightarrow{ED} = \frac{1}{2}\overrightarrow{AB} \quad \because \overrightarrow{AB} = \underline{b} - \underline{a}$$

$$\text{Also } \overrightarrow{ED} \parallel \overrightarrow{AB}$$

$$\text{And } |\overrightarrow{ED}| = \frac{1}{2}|\overrightarrow{AB}|$$

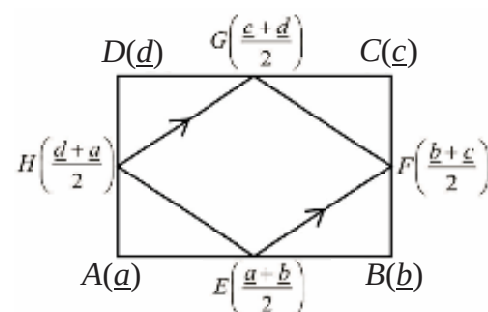
Q.14 Prove that the line segments joining the mid points of the sides of a quadrilateral taken in order form a parallelogram.

Solution:

Let $A(\underline{a}), B(\underline{b}), C(\underline{c})$ and $D(\underline{d})$ be the vertices of quadrilateral $ABCD$ respectively. E, F, G and H are mid points of $\overline{AB}, \overline{BC}, \overline{CD}$ and $\overline{DA}$ respectively.

$$E\left(\frac{\underline{a}+\underline{b}}{2}\right), \quad F\left(\frac{\underline{b}+\underline{c}}{2}\right)$$

$$G\left(\frac{\underline{c}+\underline{d}}{2}\right) \text{ and } H\left(\frac{\underline{d}+\underline{a}}{2}\right)$$



$$\begin{aligned} \overrightarrow{EF} &= \left(\frac{\underline{b}+\underline{c}}{2}\right) - \left(\frac{\underline{a}+\underline{b}}{2}\right) \\ &= \frac{\underline{b}+\underline{c}-\underline{a}-\underline{b}}{2} \end{aligned}$$

$$\overrightarrow{EF} = \frac{\underline{c}-\underline{a}}{2} \dots (i)$$

$$\begin{aligned} \overrightarrow{HG} &= \left(\frac{\underline{c}+\underline{d}}{2}\right) - \left(\frac{\underline{d}+\underline{a}}{2}\right) \\ &= \frac{\underline{c}+\underline{d}-\underline{d}-\underline{a}}{2} \end{aligned}$$

$$\overrightarrow{HG} = \frac{\underline{c}-\underline{a}}{2} \dots (ii)$$

From (i) and (ii)

$$\text{We have } \overrightarrow{EF} = \overrightarrow{HG}$$

$$\text{Also } \overrightarrow{FG} = \overrightarrow{EH}$$

Hence $EFGH$ form a parallelogram.

Vectors in Space:

The set $R^3 = \{(x, y, z) : x, y, z \in R\}$ is called the 3-dimensional space.

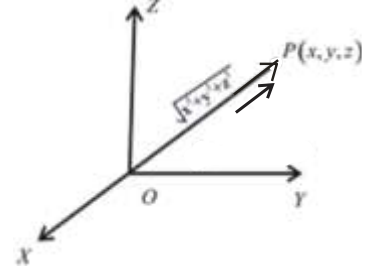
(i) Position Vector

The position vector of a point $P(x, y, z)$ in space, from the origin $O(0, 0, 0)$ is

$$\overrightarrow{OP} = x\underline{i} + y\underline{j} + z\underline{k}$$

The magnitude of $\overrightarrow{OP}$ is the distance of point P from the origin, i.e.

$$|\overrightarrow{OP}| = \sqrt{x^2 + y^2 + z^2}$$

**(ii) The unit vectors $\underline{i}, \underline{j}, \underline{k}$**

$\underline{i}, \underline{j}$ and $\underline{k}$ are called unit vectors along X, Y, Z axes respectively. They are written as:

$$\underline{i} = [1, 0, 0] \quad \underline{j} = [0, 1, 0] \quad \underline{k} = [0, 0, 1]$$

$$|\underline{i}| = \sqrt{(1)^2 + (0)^2 + (0)^2} = 1$$

$$|\underline{j}| = \sqrt{(0)^2 + (1)^2 + (0)^2} = 1$$

$$|\underline{k}| = \sqrt{(0)^2 + (0)^2 + (1)^2} = 1$$

A vector $\underline{v}$ can be written as:

$$\underline{v} = [x, y, z] = [x, 0, 0] + [0, y, 0] + [0, 0, z]$$

$$= x[1, 0, 0] + y[0, 1, 0] + z[0, 0, 1]$$

$$= x\underline{i} + y\underline{j} + z\underline{k}$$

Similarly, sum of two vectors $\underline{u}$ and $\underline{v}$ can be written as:

$$\underline{u} = [x_1, y_1, z_1], \quad \underline{v} = [x_2, y_2, z_2]$$

$$\underline{u} + \underline{v} = [x_1 + x_2, y_1 + y_2, z_1 + z_2]$$

$$= (x_1 + x_2)\underline{i} + (y_1 + y_2)\underline{j} + (z_1 + z_2)\underline{k}$$

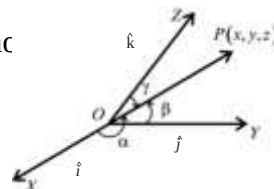
(iii) Distance between two points in space:

The distance between two points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$ in space is given by:

$$|\overline{AB}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

(iv) Direction Angles and Direction Cosines of a vector:

Let $\underline{r} = \overline{OP} = x\underline{i} + y\underline{j} + z\underline{k}$ be a non-zero vector, let α, β and γ denoted between $\underline{r}$ and the unit coordinate vector $\underline{i}, \underline{j}$ and $\underline{k}$ respectively, $0 \leq \beta \leq \pi$ and $0 \leq \gamma \leq \pi$.



(i) The angles α, β, γ are called the direction angles.

(ii) The numbers $\cos \alpha, \cos \beta$ and $\cos \gamma$ are called direction cosines.

Important Result:

Prove that

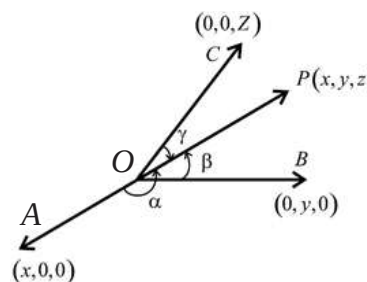
$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

Proof:

Let $\underline{r} = [x, y, z] = x\underline{i} + y\underline{j} + z\underline{k}$

$$\therefore |\underline{r}| = \sqrt{x^2 + y^2 + z^2} = r$$

Then $\frac{\underline{r}}{|\underline{r}|} = \left[\frac{x}{r}, \frac{y}{r}, \frac{z}{r} \right]$ is the unit vector in the direction of the vector $\underline{r} = \overline{OP}$. It can be



visualized that the triangle OAP is a right triangle with $\angle A = 90^\circ$. Therefore in right triangle OAP ,

$$\cos \alpha = \frac{\overline{OA}}{\overline{OP}} = \frac{x}{r},$$

Similarly

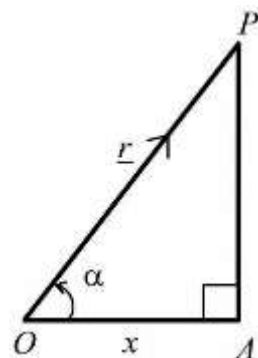
$$\cos \beta = \frac{y}{r}, \quad \cos \gamma = \frac{z}{r}$$

The numbers $\cos \alpha = \frac{x}{r}$, $\cos \beta = \frac{y}{r}$ and $\cos \gamma = \frac{z}{r}$ are called the

direction cosines of $\overline{OP}$.

$$\begin{aligned} \therefore \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma &= \frac{x^2}{r^2} + \frac{y^2}{r^2} + \frac{z^2}{r^2} \\ &= \frac{x^2 + y^2 + z^2}{r^2} = \frac{r^2}{r^2} \end{aligned}$$

$$\boxed{\therefore \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1}$$



EXERCISE 7.2

Q.1 Let $A = (2, 5)$ $B = (-1, 1)$ and $C = (2, -6)$. Find

(i) $\overrightarrow{AB}$

Solution:

$$\begin{aligned}\overrightarrow{AB} &= (-\underline{i} + \underline{j}) - (2\underline{i} + 5\underline{j}) \\ &= (-1-2)\underline{i} + (1-5)\underline{j}\end{aligned}$$

$$\boxed{\overrightarrow{AB} = -3\underline{i} - 4\underline{j}}$$

(ii) $2\overrightarrow{AB} - \overrightarrow{CB}$

Solution:

$$\begin{aligned}\overrightarrow{AB} &= (-\underline{i} + \underline{j}) - (2\underline{i} + 5\underline{j}) \\ &= -3\underline{i} - 4\underline{j}\end{aligned}$$

$$\begin{aligned}\overrightarrow{CB} &= (-\underline{i} + \underline{j}) - (2\underline{i} - 6\underline{j}) \\ &= -3\underline{i} + 7\underline{j}\end{aligned}$$

$$\begin{aligned}2\overrightarrow{AB} - \overrightarrow{CB} &= 2(-3\underline{i} - 4\underline{j}) - (-3\underline{i} + 7\underline{j}) \\ &= (-6\underline{i} - 8\underline{j}) - (-3\underline{i} + 7\underline{j}) \\ &= (-6+3)\underline{i} + (-8-7)\underline{j}\end{aligned}$$

$$\boxed{2\overrightarrow{AB} - \overrightarrow{CB} = -3\underline{i} - 15\underline{j}}$$

(iii) $2\overrightarrow{CB} - 2\overrightarrow{CA}$

Solution:

$$\begin{aligned}\overrightarrow{CB} &= (-\underline{i} + \underline{j}) - (2\underline{i} - 6\underline{j}) \\ &= -3\underline{i} + 7\underline{j}\end{aligned}$$

$$\begin{aligned}\overrightarrow{CA} &= (2\underline{i} + 5\underline{j}) - (2\underline{i} - 6\underline{j}) \\ &= (2-2)\underline{i} + (5+6)\underline{j} \\ &= 0\underline{i} + 11\underline{j}\end{aligned}$$

$$\begin{aligned}2\overrightarrow{CB} - 2\overrightarrow{CA} &= 2(-3\underline{i} + 7\underline{j}) - 2(0\underline{i} + 11\underline{j}) \\ &= (-6\underline{i} + 14\underline{j}) - (0\underline{i} + 22\underline{j})\end{aligned}$$

$$\boxed{2\overrightarrow{CB} - 2\overrightarrow{CA} = -6\underline{i} - 8\underline{j}}$$

Q.2 Let $\underline{u} = \underline{i} + 2\underline{j} - \underline{k}$, $\underline{v} = 3\underline{i} - 2\underline{j} + 2\underline{k}$ $\underline{w} = 5\underline{i} - \underline{j} + 3\underline{k}$. Find the indicated vector or number.

(i) $\underline{u} + 2\underline{v} + \underline{w}$

Solution:

$$\begin{aligned}\underline{u} + 2\underline{v} + \underline{w} &= (\underline{i} + 2\underline{j} - \underline{k}) + 2(3\underline{i} - 2\underline{j} + 2\underline{k}) \\ &\quad + (5\underline{i} - \underline{j} + 3\underline{k}) \\ &= \underline{i} + 2\underline{j} - \underline{k} + 6\underline{i} - 4\underline{j} + 4\underline{k} \\ &\quad + 5\underline{i} - \underline{j} + 3\underline{k}\end{aligned}$$

$$\boxed{\underline{u} + 2\underline{v} + \underline{w} = 12\underline{i} - 3\underline{j} + 6\underline{k}}$$

(ii) $\underline{v} - 3\underline{w}$

Solution:

$$\begin{aligned}\underline{v} - 3\underline{w} &= (3\underline{i} - 2\underline{j} + 2\underline{k}) - 3(5\underline{i} - \underline{j} + 3\underline{k}) \\ &= (3\underline{i} - 2\underline{j} + 2\underline{k}) - 15\underline{i} + 3\underline{j} - 9\underline{k}\end{aligned}$$

$$\boxed{\underline{v} - 3\underline{w} = -12\underline{i} + \underline{j} - 7\underline{k}}$$

(iii) $|3\underline{v} + \underline{w}|$

Solution:

$$\begin{aligned}3\underline{v} + \underline{w} &= 3(3\underline{i} - 2\underline{j} + 2\underline{k}) + (5\underline{i} - \underline{j} + 3\underline{k}) \\ &= 9\underline{i} - 6\underline{j} + 6\underline{k} + 5\underline{i} - \underline{j} + 3\underline{k} \\ &= 14\underline{i} - 7\underline{j} + 9\underline{k}\end{aligned}$$

$$|3\underline{v} + \underline{w}| = |14\underline{i} + (-7)\underline{j} + 9\underline{k}|$$

$$= \sqrt{(14)^2 + (-7)^2 + (9)^2}$$

$$= \sqrt{196 + 49 + 81}$$

$$\boxed{|3\underline{v} + \underline{w}| = \sqrt{326}}$$

Q.3 Find the magnitude of the vector $\underline{v}$ and write the direction cosines of $\underline{v}$.

(i) $\underline{v} = 2\underline{i} + 3\underline{j} + 4\underline{k}$

Solution:

$$\underline{v} = 2\underline{i} + 3\underline{j} + 4\underline{k}$$

$$|\underline{v}| = \sqrt{(2)^2 + (3)^2 + (4)^2}$$

$$= \sqrt{4+9+16}$$

$$= \sqrt{29}$$

Thus the direction cosines of $\underline{v}$ are

$$\left[\frac{2}{\sqrt{29}}, \frac{3}{\sqrt{29}}, \frac{4}{\sqrt{29}} \right]$$

(ii) $\underline{v} = \underline{i} - \underline{j} - \underline{k}$

Solution:

$$\begin{aligned} |\underline{v}| &= |\underline{i} - \underline{j} - \underline{k}| \\ &= \sqrt{(1)^2 + (1)^2 + (1)^2} \\ &= \sqrt{3} \end{aligned}$$

Thus the direction cosines of $\underline{v}$ are

$$\left[\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{-1}{\sqrt{3}} \right]$$

(iii) $\underline{v} = 4\underline{i} - 5\underline{j}$

Solution:

$$\begin{aligned} |\underline{v}| &= |4\underline{i} - 5\underline{j}| \\ &= \sqrt{(4)^2 + (5)^2} \\ &= \sqrt{16+25} \\ &= \sqrt{41} \end{aligned}$$

Thus the direction cosines of $\underline{v}$ are

$$\left[\frac{4}{\sqrt{41}}, \frac{-5}{\sqrt{41}}, 0 \right]$$

Q.4 Find α so that

$$|\alpha \underline{i} + (\alpha+1) \underline{j} + 2\underline{k}| = 3$$

Solution:

$$\begin{aligned} |\alpha \underline{i} + (\alpha+1) \underline{j} + 2\underline{k}| &= 3 \\ \sqrt{\alpha^2 + (\alpha+1)^2 + 2^2} &= 3 \end{aligned}$$

Taking square on both sides

$$\alpha^2 + (\alpha+1)^2 + 4 = 9$$

$$\alpha^2 + \alpha^2 + 1 + 2\alpha + 4 - 9 = 0$$

$$2\alpha^2 + 2\alpha - 4 = 0$$

$$2(\alpha^2 + \alpha - 2) = 0$$

$$\alpha^2 + \alpha - 2 = 0$$

$$\alpha^2 + 2\alpha - \alpha - 2 = 0$$

$$\alpha(\alpha+2) - 1(\alpha+2) = 0$$

$$(\alpha-1)(\alpha+2) = 0$$

$$\text{Either } \alpha-1=0 \quad \text{Or } \alpha+2=0$$

$$\boxed{\alpha=1}$$

$$\boxed{\alpha=-2}$$

Q.5 Find a unit vector in the direction of $\underline{v} = \underline{i} + 2\underline{j} - \underline{k}$

Solution:

Let $\hat{\underline{v}}$ be unit vector in the direction

$$\text{of } \underline{v} \text{ then } \hat{\underline{v}} = \frac{\underline{v}}{|\underline{v}|}$$

$$\underline{v} = \underline{i} + 2\underline{j} - \underline{k}$$

$$\begin{aligned} |\underline{v}| &= |\underline{i} + 2\underline{j} - \underline{k}| \\ &= \sqrt{1^2 + 2^2 + (-1)^2} \\ &= \sqrt{6} \end{aligned}$$

$$\hat{\underline{v}} = \frac{\underline{v}}{|\underline{v}|} = \frac{\underline{i} + 2\underline{j} - \underline{k}}{\sqrt{6}}$$

$$\hat{\underline{v}} = \frac{1}{\sqrt{6}} \underline{i} + \frac{2}{\sqrt{6}} \underline{j} - \frac{1}{\sqrt{6}} \underline{k}$$

Q.6 If $\underline{a} = 3\underline{i} - \underline{j} - 4\underline{k}$, $\underline{b} = -2\underline{i} - 4\underline{j} - 3\underline{k}$ and $\underline{c} = \underline{i} + 2\underline{j} - \underline{k}$. Find the unit vector parallel to $3\underline{a} - 2\underline{b} + 4\underline{c}$.

Solution:

$$\text{Let } \underline{v} = 3\underline{a} - 2\underline{b} + 4\underline{c}$$

$$\begin{aligned} &= 3(3\underline{i} - \underline{j} - 4\underline{k}) - 2(-2\underline{i} - 4\underline{j} - 3\underline{k}) \\ &\quad + 4(\underline{i} + 2\underline{j} - \underline{k}) \end{aligned}$$

$$= 9\underline{i} - 3\underline{j} - 12\underline{k} + 4\underline{i} + 8\underline{j} +$$

$$6\underline{k} + 4\underline{i} + 8\underline{j} - 4\underline{k}$$

$$\underline{v} = 17\underline{i} + 13\underline{j} - 10\underline{k}$$

$$\begin{aligned} |\underline{v}| &= |17\underline{i} + 13\underline{j} - 10\underline{k}| \\ &= \sqrt{(17)^2 + (13)^2 + (-10)^2} \\ &= \sqrt{289 + 169 + 100} \\ &= \sqrt{558} \end{aligned}$$

$$\begin{aligned} \hat{\underline{v}} &= \frac{\underline{v}}{|\underline{v}|} \\ &= \frac{17\underline{i} + 13\underline{j} - 10\underline{k}}{\sqrt{558}} \end{aligned}$$

$$\hat{\underline{v}} = \frac{17}{\sqrt{558}} \underline{i} + \frac{13}{\sqrt{558}} \underline{j} - \frac{10}{\sqrt{558}} \underline{k}$$

Q.7 Find a vector whose

- (i) **Magnitude is 4 and is parallel to**
 $2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$

Solution:

$$\text{Let } \mathbf{v} = 2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$$

$$\begin{aligned} |\mathbf{v}| &= |2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}| \\ &= \sqrt{2^2 + (-3)^2 + (6)^2} \\ &= \sqrt{4 + 9 + 36} \\ &= \sqrt{49} \\ &= 7 \end{aligned}$$

If $\hat{\mathbf{v}}$ is the unit vector in the direction of $\mathbf{v}$, then

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}}{7}$$

Thus the vector

$$4\hat{\mathbf{v}} = 4 \left(\frac{2\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}}{7} \right)$$

$$\boxed{4\hat{\mathbf{v}} = \frac{8}{7}\mathbf{i} - \frac{12}{7}\mathbf{j} + \frac{24}{7}\mathbf{k}}$$

- (ii) **Magnitude is 2 and is parallel to**
 $-\mathbf{i} + \mathbf{j} + \mathbf{k}$

Solution:

$$\text{Let } \mathbf{v} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$$

$$\begin{aligned} |\mathbf{v}| &= |-\mathbf{i} + \mathbf{j} + \mathbf{k}| \\ &= \sqrt{(-1)^2 + (1)^2 + (1)^2} \\ &= \sqrt{3} \end{aligned}$$

If $\hat{\mathbf{v}}$ is unit vector in the direction of $\mathbf{v}$ then ,

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{|\mathbf{v}|} = \frac{-\mathbf{i} + \mathbf{j} + \mathbf{k}}{\sqrt{3}}$$

Thus the vector

$$2\hat{\mathbf{v}} = 2 \left(\frac{-\mathbf{i} + \mathbf{j} + \mathbf{k}}{\sqrt{3}} \right)$$

$$\boxed{2\hat{\mathbf{v}} = \frac{-2}{\sqrt{3}}\mathbf{i} + \frac{2}{\sqrt{3}}\mathbf{j} + \frac{2}{\sqrt{3}}\mathbf{k}}$$

- Q.8 If** $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k}$, $\mathbf{v} = -\mathbf{i} + 3\mathbf{j} - \mathbf{k}$
 and $\mathbf{w} = \mathbf{i} + 6\mathbf{j} + 7\mathbf{k}$ **represent the**
sides of a triangle, find the value of
 $\mathbf{z}$.

Solution:

Property of triangle in vector.

$$\mathbf{u} + \mathbf{v} = \mathbf{w}$$

$$\begin{aligned} 2\mathbf{i} + 3\mathbf{j} + 4\mathbf{k} + (-\mathbf{i} + 3\mathbf{j} - \mathbf{k}) \\ = \mathbf{i} + 6\mathbf{j} + 3\mathbf{k} \end{aligned}$$

$$\mathbf{i} + 6\mathbf{j} + 3\mathbf{k} = \mathbf{i} + 6\mathbf{j} + \mathbf{z}\mathbf{k}$$

$$\boxed{\mathbf{z} = 3}$$

- Q.9 The position vectors of the points**
 A, B, C, D **are** $2\mathbf{i} - \mathbf{j} + \mathbf{k}$, $3\mathbf{i} + \mathbf{j}$,
 $2\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$ **and** $-\mathbf{i} - 2\mathbf{j} + \mathbf{k}$
respectively. Show that $\overrightarrow{AB}$ **is**
parallel to $\overrightarrow{CD}$.

Solution:

$$\begin{aligned} \overrightarrow{AB} &= 3\mathbf{i} + \mathbf{j} - (2\mathbf{i} - \mathbf{j} + \mathbf{k}) \\ &= \mathbf{i} + 2\mathbf{j} - \mathbf{k} \end{aligned}$$

$$\begin{aligned} \overrightarrow{CD} &= (-\mathbf{i} - 2\mathbf{j} + \mathbf{k}) - (2\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}) \\ &= -3\mathbf{i} - 6\mathbf{j} + 3\mathbf{k} \\ &= -3(\mathbf{i} + 2\mathbf{j} - \mathbf{k}) \end{aligned}$$

$$\overrightarrow{CD} = -3 \overrightarrow{AB}$$

Hence $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are parallel

- Q.10 We say that two vectors** $\mathbf{v}$ **and** $\mathbf{w}$
in space are parallel if there is a
scalar c **such that** $\mathbf{v} = c\mathbf{w}$ **the**
vectors point in the same direction
if $c > 0$, **and the vectors point in the**
opposite direction if $c < 0$

- (a) **Find two vectors of length 2**
parallel to the vector

$$\mathbf{v} = 2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}$$

Solution:

$$\mathbf{v} = 2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}$$

$$\begin{aligned} |\mathbf{v}| &= |2\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}| \\ &= \sqrt{2^2 + (-4)^2 + (4)^2} \end{aligned}$$

$$\begin{aligned}
 &= \sqrt{4+16+16} \\
 &= \sqrt{36} \\
 &= 6
 \end{aligned}$$

If it is unit vector parallel to $\underline{v}$ then,

$$\begin{aligned}
 \hat{\underline{v}} &= \frac{\underline{v}}{|\underline{v}|} \\
 &= \frac{2\underline{i} + 4\underline{j} + 4\underline{k}}{6} \\
 &= \frac{2}{6}\underline{i} - \frac{4}{6}\underline{j} + \frac{4}{6}\underline{k} \\
 &= \frac{1}{3}\underline{i} - \frac{2}{3}\underline{j} + \frac{2}{3}\underline{k}
 \end{aligned}$$

The two vectors of length 2 and parallel to $\hat{\underline{v}}$ are $2\hat{\underline{v}}$ and $-2\hat{\underline{v}}$

$$2\hat{\underline{v}} = 2\left(\frac{1}{3}\underline{i} - \frac{2}{3}\underline{j} + \frac{2}{3}\underline{k}\right)$$

and

$$-2\hat{\underline{v}} = -2\left(\frac{1}{3}\underline{i} - \frac{2}{3}\underline{j} + \frac{2}{3}\underline{k}\right)$$

$$2\hat{\underline{v}} = \frac{2}{3}\underline{i} - \frac{4}{3}\underline{j} + \frac{4}{3}\underline{k}$$

and

$$-2\hat{\underline{v}} = -\frac{2}{3}\underline{i} + \frac{4}{3}\underline{j} - \frac{4}{3}\underline{k}$$

- (b) Find the constant α so that the vectors $\underline{v} = \underline{i} - 3\underline{j} + 4\underline{k}$ and $\underline{w} = \alpha\underline{i} + 9\underline{j} - 12\underline{k}$ are parallel.

Solution:

If $\underline{v}$ and $\underline{w}$ are parallel then, $\underline{v} = c\underline{w}$

$$\underline{i} - 3\underline{j} + 4\underline{k} = c(\alpha\underline{i} + 9\underline{j} - 12\underline{k})$$

$$\underline{i} - 3\underline{j} + 4\underline{k} = c\alpha\underline{i} + 9c\underline{j} - 12c\underline{k}$$

Comparing both side

$$\alpha c = 1 \dots (i)$$

$$9c = -3$$

$$c = -\frac{1}{3}$$

Put in (i)

$$\alpha = -3$$

- (c) Find a vector of length 5 in the direction opposite that of

$$\underline{v} = \underline{i} - 2\underline{j} + 3\underline{k}.$$

Solution:

$$\underline{v} = \underline{i} - 2\underline{j} + 3\underline{k}$$

$$\begin{aligned}
 |\underline{v}| &= |\underline{i} - 2\underline{j} + 3\underline{k}| \\
 &= \sqrt{(1)^2 + (-2)^2 + (3)^2} \\
 &= \sqrt{1+4+9} \\
 &= \sqrt{14}
 \end{aligned}$$

If $\hat{\underline{v}}$ be unit vector in the direction of $\underline{v}$ then

$$\hat{\underline{v}} = \frac{\underline{v}}{|\underline{v}|} = \frac{\underline{i} - 2\underline{j} + 3\underline{k}}{\sqrt{14}}$$

Thus the required vector of length 5 and direction opposite is:

$$\begin{aligned}
 -5\hat{\underline{v}} &= -5\left(\frac{\underline{i} - 2\underline{j} + 3\underline{k}}{\sqrt{14}}\right) \\
 &= \frac{-5}{\sqrt{14}}\underline{i} + \frac{10}{\sqrt{14}}\underline{j} - \frac{15}{\sqrt{14}}\underline{k}
 \end{aligned}$$

- (d) Find a and b so that the vectors $3\underline{i} - \underline{j} + 4\underline{k}$ and $a\underline{i} + b\underline{j} - 2\underline{k}$ are Parallel.

Solution:

Let $\underline{v} = 3\underline{i} - \underline{j} + 4\underline{k}$ and $\underline{w} = a\underline{i} + b\underline{j} - 2\underline{k}$

$\underline{v}$ and $\underline{w}$ are parallel if

$$\underline{w} = c\underline{v}$$

$$a\underline{i} + b\underline{j} - 2\underline{k} = c(3\underline{i} - \underline{j} + 4\underline{k})$$

$$a\underline{i} + b\underline{j} - 2\underline{k} = 3c\underline{i} - c\underline{j} + 4c\underline{k}$$

Comparing both sides

$$a = 3c \dots (i)$$

$$b = -c \dots (ii)$$

$$-2 = 4c \dots (iii)$$

$$\Rightarrow c = -\frac{1}{2}$$

Put value of c in equation (i) and (ii), we have

$$a = 3\left(-\frac{1}{2}\right) \quad \text{and} \quad b = -\left(-\frac{1}{2}\right)$$

$$a = -\frac{3}{2}$$

$$\text{and} \quad b = \frac{1}{2}$$

Q.11 Find the direction cosines for the given vector:

(i) $\underline{v} = 3\underline{i} - \underline{j} + 2\underline{k}$

Solution:

$$\underline{v} = 3\underline{i} - \underline{j} + 2\underline{k}$$

$$|\underline{v}| = |3\underline{i} - \underline{j} + 2\underline{k}|$$

$$= \sqrt{(3)^2 + (-1)^2 + (2)^2}$$

$$= \sqrt{9+1+4}$$

$$= \sqrt{14}$$

The direction cosines of $\underline{v}$ are:

$$\left[\frac{3}{\sqrt{14}}, \frac{-1}{\sqrt{14}}, \frac{2}{\sqrt{14}} \right]$$

(ii) $6\underline{i} - 2\underline{j} + \underline{k}$

Solution:

$$\underline{v} = 6\underline{i} - 2\underline{j} + \underline{k}$$

$$|\underline{v}| = |6\underline{i} - 2\underline{j} + \underline{k}|$$

$$= \sqrt{6^2 + (-2)^2 + (1)^2}$$

$$= \sqrt{36+4+1}$$

$$= \sqrt{41}$$

The direction cosines of $\underline{v}$ are:

$$\left[\frac{6}{\sqrt{41}}, \frac{-2}{\sqrt{41}}, \frac{1}{\sqrt{41}} \right]$$

(iii) $\overrightarrow{PQ}$, where $P = (2, 1, 5)$ and $Q = (1, 3, 1)$.

Solution:

$$\overrightarrow{PQ} = (\underline{i} + 3\underline{j} + \underline{k}) - (2\underline{i} + \underline{j} + 5\underline{k})$$

$$\overrightarrow{PQ} = -\hat{i} + 2\hat{j} - 4\hat{k}$$

$$|\overrightarrow{PQ}| = \sqrt{(-1)^2 + (2)^2 + (-4)^2}$$

$$= \sqrt{1+4+16}$$

$$= \sqrt{21}$$

$$|\overrightarrow{PQ}| = \sqrt{21}$$

The direction cosines of $\overrightarrow{PQ}$ are:

$$\left[\frac{-1}{\sqrt{21}}, \frac{2}{\sqrt{21}}, \frac{-4}{\sqrt{21}} \right]$$

Q.12 Which of the following triples can be the direction angles of a single vector:

(i) $45^\circ, 45^\circ, 60^\circ$

Solution:

$$45^\circ, 45^\circ, 60^\circ$$

$$\text{Let } \alpha = 45^\circ, \beta = 45^\circ, \gamma = 60^\circ$$

We are to show that

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\text{L.H.S} = \cos^2(45^\circ) + \cos^2(45^\circ) + \cos^2(60^\circ)$$

$$= \left(\frac{1}{\sqrt{2}} \right)^2 + \left(\frac{1}{\sqrt{2}} \right)^2 + \left(\frac{1}{2} \right)^2$$

$$= \frac{1}{2} + \frac{1}{2} + \frac{1}{4}$$

$$= \frac{2+2+1}{4}$$

$$= \frac{5}{4} \neq 1$$

Thus the given angles are not direction angles of a vector.

(ii) $30^\circ, 45^\circ, 60^\circ$

Solution:

Let $\alpha = 30^\circ$, $\beta = 45^\circ$, $\gamma = 60^\circ$

We are to show that

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\text{L.H.S} = \cos^2(30) + \cos^2(45^\circ) + \cos^2(60^\circ)$$

$$= \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2$$

$$= \frac{3}{4} + \frac{1}{2} + \frac{1}{4}$$

$$= \frac{3+2+1}{4}$$

$$= \frac{6}{4} \neq 1$$

Thus the given angles are not direction angle of the vector .

(iii) $45^\circ, 60^\circ, 60^\circ$

Solution:

Let $\alpha = 45^\circ$, $\beta = 60^\circ$, $\gamma = 60^\circ$

We are to show that

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\text{L.H.S} = \cos^2(45) + \cos^2(60^\circ) + \cos^2(60^\circ)$$

$$= \left(\frac{1}{\sqrt{2}}\right)^2 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2$$

$$= \frac{1}{2} + \frac{1}{4} + \frac{1}{4}$$

$$= \frac{2+1+1}{4}$$

$$= \frac{4}{4}$$

$$= 1$$

Thus given angles are direction angles of a vector.

THE SCALAR PRODUCT OF TWO VECTORS:**Definition: 1**

The scalar or dot product of two vectors $\underline{u}$ and $\underline{v}$ in a plane or in space is

$$\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta$$

where θ is the angle between $\underline{u}$ and $\underline{v}$ and $0 \leq \theta \leq \pi$.

The unit vectors $\underline{i}, \underline{j}, \underline{k}$:

$$(a) \quad \underline{i} \cdot \underline{i} = |\underline{i}| |\underline{i}| \cos 0^\circ = (1)(1)(1) = 1 \quad \text{as } \cos 0^\circ = 1$$

$$\underline{j} \cdot \underline{j} = |\underline{j}| |\underline{j}| \cos 0^\circ = (1)(1)(1) = 1$$

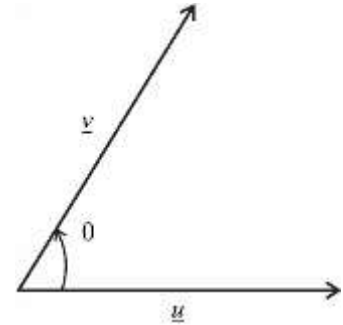
$$\underline{k} \cdot \underline{k} = |\underline{k}| |\underline{k}| \cos 0^\circ = (1)(1)(1) = 1$$

$$(b) \quad \underline{i} \cdot \underline{j} = |\underline{i}| |\underline{j}| \cos 90^\circ = (1)(1)(0) = 0 \quad \text{as } \cos 90^\circ = 0$$

$$\underline{j} \cdot \underline{k} = |\underline{j}| |\underline{k}| \cos 90^\circ = (1)(1)(0) = 0$$

$$\underline{k} \cdot \underline{i} = |\underline{k}| |\underline{i}| \cos 90^\circ = (1)(1)(0) = 0$$

$$(c) \quad \underline{u} \cdot \underline{v} = \underline{v} \cdot \underline{u} \quad (\text{Dot product of two vectors is commutative})$$

**Definition: 2**

(a) If $\underline{u} = a_1 \underline{i} + b_1 \underline{j}$ and $\underline{v} = a_2 \underline{i} + b_2 \underline{j}$ are two vectors in a plane, then the dot product of $\underline{u}$ and $\underline{v}$ is

$$\begin{aligned} \underline{u} \cdot \underline{v} &= (a_1 \underline{i} + b_1 \underline{j}) \cdot (a_2 \underline{i} + b_2 \underline{j}) \\ &= a_1 a_2 (\underline{i} \cdot \underline{i}) + a_1 b_2 (\underline{i} \cdot \underline{j}) + a_2 b_1 (\underline{j} \cdot \underline{i}) + (b_1 b_2) (\underline{j} \cdot \underline{j}) \\ &= a_1 a_2 (1) + a_1 b_2 (0) + a_2 b_1 (0) + b_1 b_2 (1) \\ \underline{u} \cdot \underline{v} &= a_1 a_2 + b_1 b_2 \end{aligned}$$

(b) If $\underline{u} = a_1 \underline{i} + b_1 \underline{j} + c_1 \underline{k}$ and $\underline{v} = a_2 \underline{i} + b_2 \underline{j} + c_2 \underline{k}$ are two vectors in space, then the dot product of $\underline{u}$ and $\underline{v}$ is

$$\begin{aligned} \underline{u} \cdot \underline{v} &= (a_1 \underline{i} + b_1 \underline{j} + c_1 \underline{k}) \cdot (a_2 \underline{i} + b_2 \underline{j} + c_2 \underline{k}) \\ &= a_1 a_2 (\underline{i} \cdot \underline{i}) + a_1 b_2 (\underline{i} \cdot \underline{j}) + a_1 c_2 (\underline{i} \cdot \underline{k}) + a_2 b_1 (\underline{j} \cdot \underline{i}) \\ &\quad + b_1 b_2 (\underline{j} \cdot \underline{j}) + b_1 c_2 (\underline{j} \cdot \underline{k}) + a_2 c_1 (\underline{k} \cdot \underline{i}) + b_2 c_1 (\underline{k} \cdot \underline{j}) + c_1 c_2 (\underline{k} \cdot \underline{k}) \\ &= a_1 a_2 (1) + a_1 b_2 (0) + a_1 c_2 (0) + a_2 b_1 (0) + b_1 b_2 (1) \\ &\quad + b_1 c_2 (0) + a_2 c_1 (0) + b_2 c_1 (0) + c_1 c_2 (1) \\ \underline{u} \cdot \underline{v} &= a_1 a_2 + b_1 b_2 + c_1 c_2 \end{aligned}$$

Perpendicular (Orthogonal) Vectors:

Two vectors $\underline{u}$ and $\underline{v}$ are perpendicular if and only if $\underline{u} \cdot \underline{v} = 0$

The angle between $\underline{u}$ and $\underline{v}$ is $\frac{\pi}{2}$

$$\therefore \underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \frac{\pi}{2} = |\underline{u}| |\underline{v}| (0)$$

$$\underline{u} \cdot \underline{v} = 0$$

Angle between two vectors:

The angles between two vectors $\underline{u}$ and $\underline{v}$ is

(a) $\underline{u} \cdot \underline{v} = |\underline{u}| |\underline{v}| \cos \theta$ where $0 \leq \theta \leq \pi$

$$\therefore \cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

(b) If $\underline{u} = a_1 \underline{i} + b_1 \underline{j} + c_1 \underline{k}$ and $\underline{v} = a_2 \underline{i} + b_2 \underline{j} + c_2 \underline{k}$

Then $\underline{u} \cdot \underline{v} = a_1 a_2 + b_1 b_2 + c_1 c_2$

$$|\underline{u}| = \sqrt{a_1^2 + b_1^2 + c_1^2} \quad |\underline{v}| = \sqrt{a_2^2 + b_2^2 + c_2^2}$$

$$\therefore \cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

$$\therefore \cos \theta = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

Projection of one vector upon another vector:

Let $\underline{u} = \overrightarrow{OA}$ and $\underline{v} = \overrightarrow{OB}$ and θ be the angle between them. Where $0 \leq \theta \leq \pi$.

Draw $BM \perp OA$. Then OM is called the projection of $\underline{v}$ along $\underline{u}$. In right triangle OMB

$$\cos \theta = \frac{\overline{OM}}{\overline{OB}}$$

$$\Rightarrow \overline{OM} = \overline{OB} \cos \theta$$

$$\therefore \overline{OM} = |\underline{v}| \cos \theta \dots (i)$$

Also $\cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|} \dots (ii)$

putting (ii) in (i)

$$\overline{OM} = |\underline{v}| \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

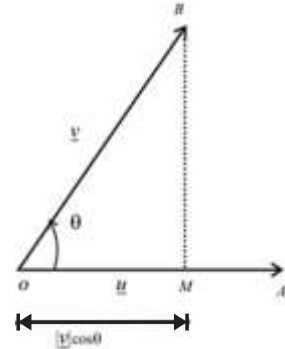
$$= \frac{\underline{u} \cdot \underline{v}}{|\underline{u}|}$$

$$= \frac{|\underline{u}| |\underline{v}| \cos \theta}{|\underline{u}|} = |\underline{v}| \cos \theta$$

i.e. projection of $\underline{v}$ along $\underline{u} = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}|} = |\underline{v}| \cos \theta$

Similarly.

Projection of $\underline{u}$ along $\underline{v} = \frac{\underline{u} \cdot \underline{v}}{|\underline{v}|} = |\underline{u}| \cos \theta$.



EXERCISE 7.3

Q.1 Find the cosine of the angle θ between $\underline{u}$ and $\underline{v}$

(i) $\underline{u} = 3\underline{i} + \underline{j} - \underline{k}$, $\underline{v} = 2\underline{i} - \underline{j} + \underline{k}$

Solution:

$$\underline{u} = 3\underline{i} + \underline{j} - \underline{k}, \underline{v} = 2\underline{i} - \underline{j} + \underline{k}$$

$$\cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

$$\begin{aligned} \cos \theta &= \frac{(3\underline{i} + \underline{j} - \underline{k}) \cdot (2\underline{i} - \underline{j} + \underline{k})}{\sqrt{3^2 + (1)^2 + (-1)^2} \sqrt{(2)^2 + (1)^2 + (1)^2}} \\ &= \frac{3(2) + (1)(-1) + (-1)(1)}{\sqrt{9+1+1} \sqrt{4+1+1}} \\ &= \frac{6-1-1}{\sqrt{11} \sqrt{6}} \\ &= \frac{4}{\sqrt{11} \sqrt{6}} \end{aligned}$$

$$\boxed{\cos \theta = \frac{4}{\sqrt{66}}}$$

(ii) $\underline{u} = \underline{i} - 3\underline{j} + 4\underline{k}$, $\underline{v} = 4\underline{i} - \underline{j} + 3\underline{k}$

Solution:

$$\underline{u} = \underline{i} - 3\underline{j} + 4\underline{k}, \underline{v} = 4\underline{i} - \underline{j} + 3\underline{k}$$

$$\cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

$$\begin{aligned} \underline{u} \cdot \underline{v} &= \frac{(\underline{i} - 3\underline{j} + 4\underline{k}) \cdot (4\underline{i} - \underline{j} + 3\underline{k})}{\sqrt{(1)^2 + (-3)^2 + (4)^2} \sqrt{(4)^2 + (-1)^2 + (3)^2}} \\ &= \frac{(1)(4) + (-3)(-1) + (4)(3)}{\sqrt{1+9+16} \sqrt{16+1+9}} \\ &= \frac{4+3+12}{\sqrt{26} \sqrt{26}} \end{aligned}$$

$$\boxed{\cos \theta = \frac{19}{\sqrt{26} \sqrt{26}} = \frac{19}{26}}$$

(iii) $\underline{u} = [-3, 5]$, $\underline{v} = [6, -2]$

Solution:

$$\underline{u} = -3\underline{i} + 5\underline{j}, \underline{v} = 6\underline{i} - 2\underline{j}$$

$$\cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

$$\begin{aligned} \cos \theta &= \frac{(-3\underline{i} + 5\underline{j}) \cdot (6\underline{i} - 2\underline{j})}{\sqrt{(-3)^2 + (5)^2} \sqrt{(6)^2 + (-2)^2}} \\ &= \frac{-18-10}{\sqrt{9+25} \sqrt{36+4}} \\ &= \frac{-28}{\sqrt{34} \sqrt{40}} \\ &= \frac{-28}{\sqrt{34 \times 40}} \\ &= \frac{-28}{\sqrt{1360}} \\ &= \frac{-28}{\sqrt{16 \times 85}} \\ &= \frac{-28}{4\sqrt{85}} \end{aligned}$$

$$\boxed{\cos \theta = \frac{-7}{\sqrt{85}}}$$

(iv) $\underline{u} = [2, -3, 1]$, $\underline{v} = [2, 4, 1]$

Solution:

$$\underline{u} = 2\underline{i} - 3\underline{j} + \underline{k}, \underline{v} = 2\underline{i} + 4\underline{j} + \underline{k}$$

$$\cos \theta = \frac{\underline{u} \cdot \underline{v}}{|\underline{u}| |\underline{v}|}$$

$$\begin{aligned} &= \frac{(2\underline{i} - 3\underline{j} + \underline{k}) \cdot (2\underline{i} + 4\underline{j} + \underline{k})}{\sqrt{(2)^2 + (-3)^2 + (1)^2} \sqrt{(2)^2 + (4)^2 + (1)^2}} \\ &= \frac{2(2) + -3(4) + (1)(1)}{\sqrt{4+9+1} \sqrt{4+16+1}} \\ &= \frac{4-12+1}{\sqrt{14} \sqrt{21}} \\ &= \frac{-7}{\sqrt{14} \sqrt{21}} \\ &= \frac{-7}{7\sqrt{6}} \end{aligned}$$

$$\boxed{\cos \theta = \frac{-1}{\sqrt{6}}}$$

Q.2 Calculate the projection of $\underline{a}$ along $\underline{b}$ and projection of $\underline{b}$ along $\underline{a}$ when

(i) $\underline{a} = \underline{i} - \underline{k}$, $\underline{b} = \underline{j} + \underline{k}$

Solution:

$$\underline{a} = \underline{i} - \underline{k} \Rightarrow |\underline{a}| = \sqrt{1+1} = \sqrt{2}$$

$$\underline{b} = \underline{j} + \underline{k} \Rightarrow |\underline{b}| = \sqrt{1+1} = \sqrt{2}$$

$$\text{Projection of } \underline{a} \text{ along } \underline{b} = \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|}$$

$$= \frac{(\underline{i} - \underline{k}) \cdot (\underline{j} + \underline{k})}{\sqrt{2}}$$

$$= \frac{0+0-1}{\sqrt{2}} = \frac{-1}{\sqrt{2}}$$

$$\text{Projection of } \underline{b} \text{ along } \underline{a} = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}|} = \frac{-1}{\sqrt{2}}$$

(ii) $\underline{a} = 3\underline{i} + \underline{j} - \underline{k}$, $\underline{b} = -2\underline{i} - \underline{j} + \underline{k}$

Solution:

$$\underline{a} = 3\underline{i} + \underline{j} - \underline{k}$$

$$|\underline{a}| = \sqrt{(3)^2 + (1)^2 + (-1)^2} = \sqrt{11}$$

$$\underline{b} = -2\underline{i} - \underline{j} + \underline{k}$$

$$|\underline{b}| = \sqrt{(-2)^2 + (-1)^2 + (1)^2} = \sqrt{6}$$

$$\text{Projection of } \underline{a} \text{ along } \underline{b} = \frac{\underline{a} \cdot \underline{b}}{|\underline{b}|}$$

$$= \frac{(3\underline{i} + \underline{j} - \underline{k}) \cdot (-2\underline{i} - \underline{j} + \underline{k})}{\sqrt{6}}$$

$$= \frac{(3)(-2) + (1)(-1) + (-1)(1)}{\sqrt{6}}$$

$$= \frac{-8}{\sqrt{6}}$$

$$\text{Projection of } \underline{b} \text{ along } \underline{a} = \frac{\underline{a} \cdot \underline{b}}{|\underline{a}|} = \frac{-8}{\sqrt{11}}$$

Q.3 Find a real number α so that the vectors $\underline{u}$ and $\underline{v}$ are perpendicular.

(i) $\underline{u} = 2\alpha\underline{i} + \underline{j} - \underline{k}$, $\underline{v} = \underline{i} + \alpha\underline{j} + 4\underline{k}$

Solution:

The vectors $\underline{u}$ and $\underline{v}$ are perpendicular

$$\text{So } \underline{u} \cdot \underline{v} = 0$$

$$(2\alpha\underline{i} + \underline{j} - \underline{k}) \cdot (\underline{i} + \alpha\underline{j} + 4\underline{k}) = 0$$

$$2\alpha + \alpha - 4 = 0$$

$$3\alpha - 4 = 0$$

$$\boxed{\alpha = \frac{4}{3}}$$

(ii) $\underline{u} = \alpha\underline{i} + 2\alpha\underline{j} - \underline{k}$, $\underline{v} = \underline{i} + \alpha\underline{j} + 3\underline{k}$

Solution:

The vectors $\underline{u}$ and $\underline{v}$ are perpendicular

$$\text{So } \underline{u} \cdot \underline{v} = 0$$

$$(\alpha\underline{i} + 2\alpha\underline{j} - \underline{k}) \cdot (\underline{i} + \alpha\underline{j} + 3\underline{k}) = 0$$

$$\alpha + 2\alpha^2 - 3 = 0$$

$$2\alpha^2 + \alpha - 3 = 0$$

$$2\alpha^2 + 3\alpha - 2\alpha - 3 = 0$$

$$\alpha(2\alpha + 3) - 1(2\alpha + 3) = 0$$

$$(\alpha - 1)(2\alpha + 3) = 0$$

$$\text{Either } \alpha - 1 = 0 \text{ or } 2\alpha + 3 = 0$$

$$\boxed{\alpha = 1}$$

$$\text{or } \boxed{\alpha = \frac{-3}{2}}$$

Q.4 Find the number z so that the triangle with vertices

$$A(1, -1, 0), B(-2, 2, 1) \text{ and}$$

$C(0, 2, z)$ is right triangle with right angle at C .

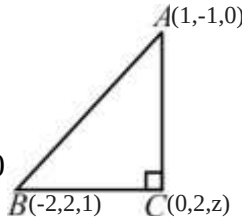
Solution:

$$\begin{aligned} \overrightarrow{AC} &= (0\underline{i} + 2\underline{j} + z\underline{k}) - (\underline{i} - \underline{j} + 0\underline{k}) \\ &= (-\underline{i} + 3\underline{j} + z\underline{k}) \end{aligned}$$

$$\begin{aligned} \overrightarrow{BC} &= (0\underline{i} + 2\underline{j} + z\underline{k}) - (-2\underline{i} + 2\underline{j} + \underline{k}) \\ &= (2\underline{i} + 0\underline{j} + (z-1)\underline{k}) \end{aligned}$$

$$\text{As } \overrightarrow{AC} \cdot \overrightarrow{BC} = 0$$

$$\begin{aligned}
 (-\underline{i} + 3\underline{j} + z\underline{k}) \cdot (2\underline{i} + 0\underline{j} + (z-1)\underline{k}) &= 0 \\
 -2 + 0 + z(z-1) &= 0 \\
 -2 + z^2 - z &= 0 \\
 z^2 - z - 2 &= 0 \\
 z^2 - 2z + z - 2 &= 0 \\
 z(z-2) + 1(z-2) &= 0 \\
 (z+1)(z-2) &= 0 \\
 \text{Either } z+1=0 &\text{ or } z-2=0 \\
 \boxed{z=-1} &\text{ or } \boxed{z=2}
 \end{aligned}$$



Q.5 If $\underline{v}$ is a vector for which $\underline{v} \cdot \underline{i} = 0, \underline{v} \cdot \underline{j} = 0, \underline{v} \cdot \underline{k} = 0$, find $\underline{v}$.

Solution:

$$\begin{aligned}
 \text{Let } \underline{v} &= x\underline{i} + y\underline{j} + z\underline{k} \dots (i) \\
 \underline{v} \cdot \underline{i} &= (x\underline{i} + y\underline{j} + z\underline{k}) \cdot \underline{i} \\
 0 &= x(\underline{i} \cdot \underline{i}) \quad \because \underline{i} \cdot \underline{i} = 1 \\
 0 &= x \\
 \underline{v} \cdot \underline{j} &= (x\underline{i} + y\underline{j} + z\underline{k}) \cdot \underline{j} \\
 0 &= y(\underline{j} \cdot \underline{j}) \quad \because \underline{j} \cdot \underline{j} = 1 \\
 0 &= y \\
 \underline{v} \cdot \underline{k} &= (x\underline{i} + y\underline{j} + z\underline{k}) \cdot \underline{k} \\
 0 &= z(\underline{k} \cdot \underline{k}) \quad \because \underline{k} \cdot \underline{k} = 1 \\
 0 &= z \\
 \text{Put the value of } x, y, z &\text{ in (i)} \\
 \underline{v} &= x\underline{i} + y\underline{j} + z\underline{k} \\
 \boxed{\underline{v} = 0\underline{i} + 0\underline{j} + 0\underline{k}}
 \end{aligned}$$

Q.6

(i) Show that the vectors $3\underline{i} - 2\underline{j} + \underline{k}$, $\underline{i} - 3\underline{j} + 5\underline{k}$ and $2\underline{i} + \underline{j} - 4\underline{k}$ form a right triangle.

Solution:

$$\begin{aligned}
 \text{Let } \underline{u} &= (3\underline{i} - 2\underline{j} + \underline{k}), \underline{v} = (\underline{i} - 3\underline{j} + 5\underline{k}) \\
 \text{and } \underline{w} &= (2\underline{i} + \underline{j} - 4\underline{k}) \text{ are vectors} \\
 \text{along sides of the triangle,} \\
 \underline{u} \cdot \underline{w} &= (3\underline{i} - 2\underline{j} + \underline{k}) \cdot (2\underline{i} + \underline{j} - 4\underline{k}) \\
 &= 3(2) - 2(1) + 1(-4) \\
 &= 6 - 2 - 4 \\
 \underline{u} \cdot \underline{w} &= 0
 \end{aligned}$$

$$\Rightarrow \underline{u} \perp \underline{w}$$

So $\underline{u}, \underline{v}$ and $\underline{w}$ form a right triangle

(ii) Show that the set of points

$$P = (1, 3, 2), Q = (4, 1, 4) \text{ and}$$

$$R = (6, 5, 5) \text{ form a right triangle.}$$

Solution:

$$\begin{aligned}
 \overrightarrow{PQ} &= (4\underline{i} + \underline{j} + 4\underline{k}) - (\underline{i} + 3\underline{j} + 2\underline{k}) \\
 &= 3\underline{i} - 2\underline{j} + 2\underline{k}
 \end{aligned}$$

$$\begin{aligned}
 \overrightarrow{QR} &= (6\underline{i} + 5\underline{j} + 5\underline{k}) - (4\underline{i} + \underline{j} + 4\underline{k}) \\
 &= 2\underline{i} + 4\underline{j} + \underline{k}
 \end{aligned}$$

$$\begin{aligned}
 \overrightarrow{PQ} \cdot \overrightarrow{QR} &= (3\underline{i} - 2\underline{j} + 2\underline{k}) \cdot (2\underline{i} + 4\underline{j} + \underline{k}) \\
 &= 6 - 8 + 2 \\
 &= 0
 \end{aligned}$$

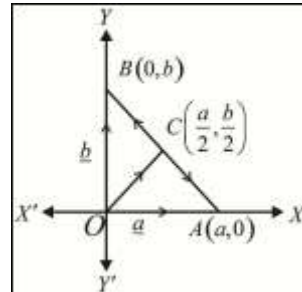
$$\Rightarrow \overrightarrow{PQ} \perp \overrightarrow{QR}$$

Thus PQR is a right triangle.

Q.7 Show that the midpoint of hypotenuse of a right triangle is equidistant from its vertices.

Solution:

Consider a right triangle AOB with "O" is origin and $A(a, 0), B(0, b)$ be its other vertices.



C be the midpoint of hypotenuse $\overline{AB}$ of a right triangle

$$\text{Then } C \left(\frac{a}{2}, \frac{b}{2} \right)$$

We have to show that

$$|\overrightarrow{OC}| = |\overrightarrow{AC}| = |\overrightarrow{BC}|$$

$$\begin{aligned}
 \overrightarrow{AC} &= \left(\frac{a}{2}\underline{i} + \frac{b}{2}\underline{j} \right) - (a\underline{i} + 0\underline{j}) \\
 &= -\frac{a}{2}\underline{i} + \frac{b}{2}\underline{j}
 \end{aligned}$$

$$\begin{aligned}
 |\overrightarrow{AC}| &= \sqrt{\left(-\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2} \\
 &= \sqrt{\frac{a^2}{4} + \frac{b^2}{4}} \\
 &= \sqrt{\frac{a^2 + b^2}{4}} \\
 &= \frac{\sqrt{a^2 + b^2}}{2} \dots (i)
 \end{aligned}$$

$$\begin{aligned}
 \overrightarrow{BC} &= \left(\frac{a}{2}\underline{i} + \frac{b}{2}\underline{j}\right) - (0\underline{i} + b\underline{j}) \\
 &= \frac{a}{2}\underline{i} - \frac{b}{2}\underline{j}
 \end{aligned}$$

$$\begin{aligned}
 |\overrightarrow{BC}| &= \sqrt{\frac{a^2}{4} + \frac{b^2}{4}} \\
 &= \frac{\sqrt{a^2 + b^2}}{2} \dots (ii)
 \end{aligned}$$

$$\begin{aligned}
 \overrightarrow{OC} &= \left(\frac{a}{2}\underline{i} + \frac{b}{2}\underline{j}\right) - (0\underline{i} + 0\underline{j}) \\
 &= \left(\frac{a}{2}\underline{i} + \frac{b}{2}\underline{j}\right)
 \end{aligned}$$

$$\begin{aligned}
 |\overrightarrow{OC}| &= \sqrt{\left(\frac{a}{2}\right)^2 + \left(\frac{b}{2}\right)^2} \\
 &= \frac{\sqrt{a^2 + b^2}}{2} \dots (iii)
 \end{aligned}$$

From (i), (ii), (iii) $|\overrightarrow{AC}| = |\overrightarrow{BC}| = |\overrightarrow{OC}|$

(The midpoint of hypotenuse of a right triangle is equidistance from its vertices)

Q.8 Prove that perpendicular bisector of the sides of a triangle are concurrent.

Solution:

Let $A(\underline{a})$, $B(\underline{b})$ and $C(\underline{c})$ be the vertices of $\triangle ABC$

D, E and F be the mid points of $\overline{AB}$, $\overline{BC}$ and $\overline{CA}$ respectively.

$$\overrightarrow{OD}\left(\frac{\underline{a} + \underline{b}}{2}\right), \overrightarrow{OE}\left(\frac{\underline{b} + \underline{c}}{2}\right) \text{ and}$$

$$\overrightarrow{OF}\left(\frac{\underline{c} + \underline{a}}{2}\right) \text{ are right}$$

bisector of sides $\overline{AB}$, $\overline{BC}$ and $\overline{AC}$ respectively.

$$\text{Now } \overrightarrow{AB} = \underline{b} - \underline{a}$$

$$\text{As } \overrightarrow{OD} \perp \overrightarrow{AB}$$

$$\Rightarrow \overrightarrow{OD} \cdot \overrightarrow{AB} = 0$$

$$\left(\frac{\underline{a} + \underline{b}}{2}\right) \cdot (\underline{b} - \underline{a}) = 0$$

$$(\underline{b} + \underline{a}) \cdot (\underline{b} - \underline{a}) = 0$$

$$b^2 - a^2 = 0 \dots (i)$$

Similarly $\overrightarrow{OE} \perp \overrightarrow{BC}$

$$\Rightarrow \overrightarrow{OE} \cdot \overrightarrow{BC} = 0$$

$$\left(\frac{\underline{b} + \underline{c}}{2}\right) \cdot (\underline{c} - \underline{b}) = 0$$

$$(\underline{c} + \underline{b}) \cdot (\underline{c} - \underline{b}) = 0$$

$$c^2 - b^2 = 0 \dots (ii)$$

Adding (i) and (ii)

$$b^2 - a^2 + c^2 - b^2 = 0$$

$$c^2 - a^2 = 0$$

$$(\underline{c} + \underline{a}) \cdot (\underline{c} - \underline{a}) = 0$$

$$\left(\frac{\underline{c} + \underline{a}}{2}\right) \cdot (\underline{c} - \underline{a}) = 0$$

$$\overrightarrow{OF} \cdot \overrightarrow{AC} = 0$$

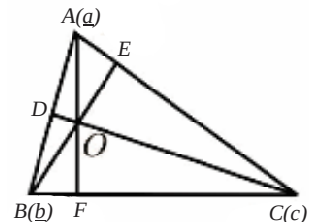
$$\overrightarrow{OF} \perp \overrightarrow{AC}$$

So the right bisectors of the sides of a triangle are concurrent.

Q.9 Prove that the altitude of a triangle are concurrent.

Solution:

Let $A(\underline{a})$, $B(\underline{b})$ and $C(\underline{c})$ be the vertices of $\triangle ABC$



$\overrightarrow{CD}, \overrightarrow{BE}$ and $\overrightarrow{AF}$ be the altitudes along sides $\overrightarrow{AB}, \overrightarrow{CA}$ and $\overrightarrow{BC}$

Now $\overrightarrow{CD} \perp \overrightarrow{AB} \Rightarrow \overrightarrow{CO} \perp \overrightarrow{AB}$

$$\therefore \overrightarrow{CO} \cdot \overrightarrow{AB} = 0$$

$$-\underline{c} \cdot (\underline{b} - \underline{a}) = 0$$

$$\Rightarrow \underline{c} \cdot (\underline{b} - \underline{a}) = 0$$

$$\underline{b} \cdot \underline{c} - \underline{a} \cdot \underline{c} = 0$$

$$\underline{b} \cdot \underline{c} = \underline{a} \cdot \underline{c} \dots (i)$$

Also $\overrightarrow{BE} \perp \overrightarrow{CA} \Rightarrow \overrightarrow{BO} \perp \overrightarrow{CA}$

$$\therefore \overrightarrow{BO} \cdot \overrightarrow{CA} = 0$$

$$-\underline{b} \cdot (\underline{a} - \underline{c}) = 0$$

$$\Rightarrow \underline{b} \cdot (\underline{a} - \underline{c}) = 0$$

$$\underline{a} \cdot \underline{b} - \underline{b} \cdot \underline{c} = 0$$

$$\underline{a} \cdot \underline{b} = \underline{b} \cdot \underline{c} \dots (ii)$$

From (i) and (ii)

$$\underline{a} \cdot \underline{b} = \underline{a} \cdot \underline{c}$$

$$\underline{a} \cdot \underline{b} - \underline{a} \cdot \underline{c} = 0$$

$$-\underline{a} \cdot (\underline{c} - \underline{b}) = 0$$

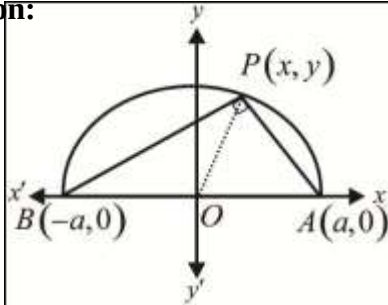
$$\overrightarrow{AO} \cdot \overrightarrow{BC} = 0$$

$$\Rightarrow \overrightarrow{AO} \perp \overrightarrow{BC} \Rightarrow \overrightarrow{AF} \perp \overrightarrow{BC}$$

Thus the altitude of a triangle are concurrent.

Q.10 Prove that the angle in a semi circle is a right angle.

Solution:



Let AOB be semicircle with radius a and centre at origin O . whereas x -axis is taken along the line AB .

Let $P(x, y)$ be any point on semicircle. Join A and B with P .

Also join O and P .

Now $\overrightarrow{OA} = a\underline{i}$, $\overrightarrow{OB} = -a\underline{i}$

$$\overrightarrow{OP} = x\underline{i} + y\underline{j} \Rightarrow |\overrightarrow{OP}| = \sqrt{x^2 + y^2}$$

$$\text{Also } |\overrightarrow{OP}| = a$$

$$\Rightarrow |\overrightarrow{OP}|^2 = a^2$$

$$x^2 + y^2 = a^2 \dots (i)$$

$$\overrightarrow{AP} = (x\underline{i} + y\underline{j}) - (a\underline{i})$$

$$= (x - a)\underline{i} + y\underline{j}$$

$$\overrightarrow{BP} = (x\underline{i} + y\underline{j}) - (-a\underline{i})$$

$$= (x + a)\underline{i} + y\underline{j}$$

$$\overrightarrow{AP} \cdot \overrightarrow{BP} = ((x - a)\underline{i} + y\underline{j}) \cdot ((x + a)\underline{i} + y\underline{j})$$

$$= (x - a)(x + a) + y^2$$

$$= x^2 - a^2 + y^2$$

$$= x^2 + y^2 - a^2$$

$$a^2 - a^2 = 0 \text{ from (i)}$$

$$\overrightarrow{AP} \cdot \overrightarrow{BP} = 0$$

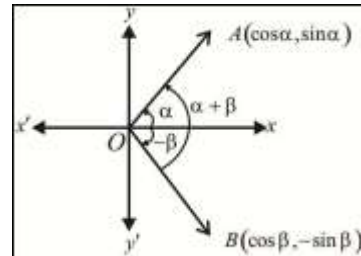
$$\overrightarrow{AP} \perp \overrightarrow{BP}$$

$$\Rightarrow m\angle APB = 90^\circ$$

Q.11 Prove that

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

Solution:



Let $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are unit vectors in xy -plane making angles α and $-\beta$ with the positive x -axis respectively.

so that $\angle AOB = \alpha + \beta$

$$\text{then } \overrightarrow{OA} = \cos \alpha \underline{i} + \sin \alpha \underline{j}$$

$$\text{and } \overrightarrow{OB} = \cos \beta \underline{i} - \sin \beta \underline{j}$$

$$\overrightarrow{OB} \cdot \overrightarrow{OA} = (\cos \beta \underline{i} - \sin \beta \underline{j}) \cdot (\cos \alpha \underline{i} + \sin \alpha \underline{j})$$

$$|\vec{OA}||\vec{OB}|\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta \quad |\vec{OA}| = |\vec{OB}| = 1$$

Q.12 Prove that in any triangle ABC .

(i) $b = c \cos A + a \cos C$

Solution:

Let the vector $\underline{a}, \underline{b}$ and $\underline{c}$ are along the sides BC, CA and AB of the triangle ABC

$$\therefore \underline{a} + \underline{b} + \underline{c} = 0$$

$$\underline{b} = -(\underline{a} + \underline{c})$$

Now taking dot

product with $\underline{b}$

$$\underline{b} \cdot \underline{b} = -(\underline{a} + \underline{c}) \cdot \underline{b}$$

$$b^2 = -(a \cdot \underline{b} + \underline{c} \cdot \underline{b})$$

$$= -\underline{a} \cdot \underline{b} - \underline{c} \cdot \underline{b}$$

$$= -ab \cos(\pi - C) - bc \cos(\pi - A)$$

$$= -ab(-\cos C) - bc(-\cos A)$$

$$b^2 = ab \cos C + bc \cos A$$

$$\boxed{b = a \cos C + c \cos A}$$

(ii) $c = a \cos B + b \cos A$

Solution:

Let the vectors $\underline{a}, \underline{b}$ and $\underline{c}$ are along the sides BC, CA and AB of the triangle ABC

$$\therefore \underline{a} + \underline{b} + \underline{c} = 0$$

$$\underline{c} = -\underline{a} - \underline{b}$$

Now taking dot product with $\underline{c}$

$$\underline{c} \cdot \underline{c} = -(\underline{a} + \underline{b}) \cdot \underline{c}$$

$$c^2 = -(\underline{a} \cdot \underline{c} + \underline{b} \cdot \underline{c})$$

$$= -\underline{a} \cdot \underline{c} - \underline{b} \cdot \underline{c}$$

$$= -ac \cos(\pi - B) - bc \cos(\pi - A)$$

$$= -ac \cos(\pi - B) - bc \cos(\pi - A)$$

$$= -ac(-\cos B) - bc(-\cos A)$$

$$c^2 = ac \cos B + bc \cos A$$

$$\boxed{c = a \cos B + b \cos A}$$

(iii) $b^2 = c^2 + a^2 - 2ca \cos B$

Solution:

Let the vectors $\underline{a}, \underline{b}$ and $\underline{c}$ along the Sides BC, CA and AB of triangle ABC .

$$\Rightarrow \underline{a} + \underline{b} + \underline{c} = 0$$

$$\underline{b} = -(\underline{a} + \underline{c})$$

$$\underline{b} \cdot \underline{b} = -(\underline{a} + \underline{c}) \cdot \underline{b}$$

$$b^2 = -(\underline{a} + \underline{c}) \cdot (-(\underline{a} + \underline{c}))$$

$$= (\underline{a} + \underline{c}) \cdot (\underline{a} + \underline{c})$$

$$= \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{c} + \underline{c} \cdot \underline{a} + \underline{c} \cdot \underline{c}$$

$$= a^2 + 2\underline{a} \cdot \underline{c} + c^2$$

$$= a^2 + c^2 + 2\underline{c} \cdot \underline{a}$$

$$= a^2 + c^2 + 2ca \cos(\pi - B)$$

$$\boxed{b^2 = a^2 + c^2 - 2ca \cos B}$$

(iv) $c^2 = a^2 + b^2 - 2ab \cos C$

Let $\underline{a}, \underline{b}$ and $\underline{c}$ are vectors along sides $\vec{BC}, \vec{CA}$ and $\vec{AB}$ of triangle ABC

$$\underline{a} + \underline{b} + \underline{c} = 0$$

$$\underline{c} = -\underline{a} - \underline{b}$$

$$\underline{c} = -(\underline{a} + \underline{b})$$

$$\underline{c} \cdot \underline{c} = -(\underline{a} + \underline{b}) \cdot (-(\underline{a} + \underline{b}))$$

$$c^2 = (\underline{a} + \underline{b}) \cdot (\underline{a} + \underline{b})$$

$$= \underline{a} \cdot \underline{a} + \underline{a} \cdot \underline{b} + \underline{b} \cdot \underline{a} + \underline{b} \cdot \underline{b}$$

$$= a^2 + 2\underline{a} \cdot \underline{b} + b^2$$

$$= a^2 + 2\underline{a} \cdot \underline{b} + b^2$$

$$= a^2 + b^2 + 2ab \cos(\pi - C)$$

$$\boxed{c^2 = a^2 + b^2 - 2ab \cos C}$$

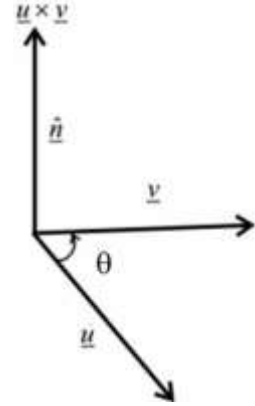
The Cross Product or Vectors Product of Two Vector:**(i) Definition 1:**

The vector or cross product of two vectors $\underline{u}$ and $\underline{v}$ in space is

$$\underline{u} \times \underline{v} = |\underline{u}| |\underline{v}| \sin \theta \underline{n}$$

where θ is the angle between $\underline{u}$ and $\underline{v}$ and $0 \leq \theta \leq \pi$. $\underline{n}$ is a unit vector perpendicular to the plane of $\underline{u}$ and $\underline{v}$ with direction given by the “right hand rule” stated below:

“If the fingers of right hand are curled in a direction from $\underline{u}$ towards $\underline{v}$, then the thumb will point in the direction of $\underline{n}$ which is $\underline{u} \times \underline{v}$. It is important to note that $\underline{u} \times \underline{v} \neq \underline{v} \times \underline{u}$, rather $\underline{u} \times \underline{v} = -(\underline{v} \times \underline{u})$.

**(ii) The Unit Vectors $\underline{i}, \underline{j}, \underline{k}$:**

$$(a) \quad \underline{i} \times \underline{i} = |\underline{i}| |\underline{i}| \sin 0^\circ \underline{n} = (1)(1)(0)\underline{n} = 0 \quad \because \sin 0^\circ = 0$$

$$\underline{j} \times \underline{j} = |\underline{j}| |\underline{j}| \sin 0^\circ \underline{n} = (1)(1)(0)\underline{n} = 0$$

$$\underline{k} \times \underline{k} = |\underline{k}| |\underline{k}| \sin 0^\circ \underline{n} = (1)(1)(0)\underline{n} = 0$$

$$(b) \quad \underline{i} \times \underline{j} = |\underline{i}| |\underline{j}| \sin 90^\circ \underline{k} = (1)(1)(1)\underline{k} = \underline{k} \quad \sin 90^\circ = 1$$

$$\underline{j} \times \underline{k} = |\underline{j}| |\underline{k}| \sin 90^\circ \underline{i} = (1)(1)(1)\underline{i} = \underline{i}$$

$$\underline{k} \times \underline{i} = |\underline{k}| |\underline{i}| \sin 90^\circ \underline{j} = (1)(1)(1)\underline{j} = \underline{j}$$

Definition 2:

If $\underline{u} = a_1 \underline{i} + b_1 \underline{j} + c_1 \underline{k}$ and $\underline{v} = a_2 \underline{i} + b_2 \underline{j} + c_2 \underline{k}$ are two vectors in space, then cross

$$\text{product of } \underline{u} \text{ and } \underline{v} \text{ is : } \underline{u} \times \underline{v} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$$

which is called the “Determinant formula” for $\underline{u} \times \underline{v}$.

Parallel Vectors:

If two vectors $\underline{u}$ and $\underline{v}$ are parallel, then

$$\underline{u} \times \underline{v} = |\underline{u}| |\underline{v}| \sin 0^\circ \underline{n} = |\underline{u}| |\underline{v}| (0) \underline{n}$$

$$\therefore \underline{u} \times \underline{v} = 0$$

Angle between two Vectors:

The angle θ between two vectors $\underline{u}$ and $\underline{v}$ is

$$\underline{u} \times \underline{v} = |\underline{u}| |\underline{v}| \sin \theta \underline{n}$$

$$\therefore \frac{\underline{u} \times \underline{v}}{\underline{n}} = |\underline{u}| |\underline{v}| \sin \theta$$

$$\therefore |\underline{u} \times \underline{v}| = |\underline{u}| |\underline{v}| \sin \theta$$

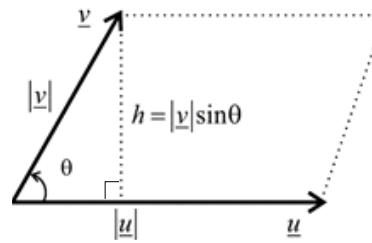
$$\therefore \sin \theta = \frac{|\underline{u} \times \underline{v}|}{|\underline{u}| |\underline{v}|}$$

Area of Parallelogram:

$$\text{Area of Parallelogram} = \text{Base} \times \text{Height}$$

$$= |\underline{u}| |\underline{v}| \sin \theta$$

$$\text{Area of Parallelogram} = |\underline{u} \times \underline{v}|$$

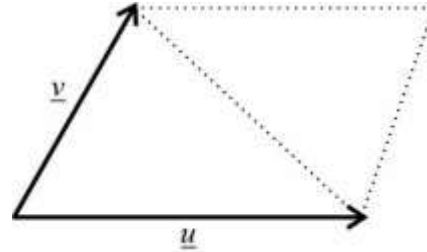
**Area of a Triangle:**

From the figure, it is clear Area of triangle

$$\text{Area of Triangle} = \frac{1}{2} (\text{Area of Parallelogram})$$

$$= \frac{1}{2} |\underline{u} \times \underline{v}| \text{ where } \underline{u} \text{ and } \underline{v} \text{ are}$$

vectors along to adjacent sides of triangle.

**EXERCISE 7.4**

Q.1 Compute the cross product $\underline{a} \times \underline{b}$ and $\underline{b} \times \underline{a}$. Check your answer by showing that each $\underline{a}$ and $\underline{b}$ is perpendicular to $\underline{a} \times \underline{b}$ and $\underline{b} \times \underline{a}$.

(i) $\underline{a} = 2\underline{i} + \underline{j} - \underline{k}$, $\underline{b} = \underline{i} - \underline{j} + \underline{k}$

Solution:

$$\underline{a} = 2\underline{i} + \underline{j} - \underline{k}, \underline{b} = \underline{i} - \underline{j} + \underline{k}$$

$$\begin{aligned} \underline{a} \times \underline{b} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{vmatrix} \\ &= \underline{i}(1-1) - \underline{j}(2+1) + \underline{k}(-2-1) \\ &= 0\underline{i} - 3\underline{j} - 3\underline{k} \end{aligned}$$

Now

$$\begin{aligned} \underline{a} \cdot (\underline{a} \times \underline{b}) &= (2\underline{i} + \underline{j} - \underline{k}) \cdot (0\underline{i} - 3\underline{j} - 3\underline{k}) \\ &= 0 - 3 + 3 \\ &= 0 \end{aligned}$$

$$\Rightarrow \underline{a} \perp \underline{a} \times \underline{b}$$

Also

$$\begin{aligned} \underline{b} \cdot (\underline{a} \times \underline{b}) &= (\underline{i} - \underline{j} + \underline{k}) \cdot (0\underline{i} - 3\underline{j} - 3\underline{k}) \\ &= (1)(0) + (-1)(-3) + (1)(-3) \\ &= 0 + 3 - 3 = 0 \end{aligned}$$

$$\Rightarrow \underline{b} \perp \underline{a} \times \underline{b}$$

Thus $\underline{a} \times \underline{b}$ is perpendicular to both vectors $\underline{a}$ and $\underline{b}$.

$$\begin{aligned} \underline{b} \times \underline{a} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{vmatrix} \\ &= \underline{i}(1-1) - \underline{j}(-1-2) + \underline{k}(1+2) \\ &= 0\underline{i} + 3\underline{j} + 3\underline{k} \end{aligned}$$

Now

$$\begin{aligned} \underline{a} \cdot (\underline{b} \times \underline{a}) &= (2\underline{i} + \underline{j} - \underline{k}) \cdot (0\underline{i} + 3\underline{j} + 3\underline{k}) \\ &= 2(0) + (1)(3) + (-1)(3) \\ &= 3 - 3 \\ &= 0 \end{aligned}$$

$$\Rightarrow \underline{a} \perp \underline{b} \times \underline{a}$$

Now

$$\begin{aligned} \underline{b} \cdot (\underline{b} \times \underline{a}) &= (\underline{i} - \underline{j} + \underline{k}) \cdot (0\underline{i} + 3\underline{j} + 3\underline{k}) \\ &= (1)(0) + (-1)(3) + (1)(3) \\ &= 0 - 3 + 3 = 0 \end{aligned}$$

Thus $\underline{b} \times \underline{a}$ is perpendicular to both the vectors $\underline{a}$ and $\underline{b}$.

(ii) $\underline{a} = \underline{i} + \underline{j}$, $\underline{b} = \underline{i} - \underline{j}$

Solution:

$$\begin{aligned}\underline{a} \times \underline{b} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 1 & 0 \\ 1 & -1 & 0 \end{vmatrix} \\ &= \underline{i}(0-0) - \underline{j}(0-0) + \underline{k}(-1-1) \\ &= -2\underline{k}\end{aligned}$$

$$\underline{a} \cdot (\underline{a} \times \underline{b}) = (\underline{i} + \underline{j}) \cdot (-2\underline{k}) = 0$$

$$\text{Also } \underline{b} \cdot (\underline{a} \times \underline{b}) = (\underline{i} - \underline{j}) \cdot (-2\underline{k}) = 0$$

Thus $\underline{a} \times \underline{b}$ is perpendicular to both vector $\underline{a}$ and $\underline{b}$.

$$\begin{aligned}\underline{b} \times \underline{a} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & -1 & 0 \\ 1 & 1 & 0 \end{vmatrix} \\ &= \underline{i}(0) - \underline{j}(0) + \underline{k}(1+1) \\ &= 2\underline{k}\end{aligned}$$

$$\underline{a} \cdot (\underline{b} \times \underline{a}) = (\underline{i} + \underline{j}) \cdot (2\underline{k})$$

$$\text{Also } \underline{b} \cdot (\underline{b} \times \underline{a}) = (\underline{i} - \underline{j}) \cdot (2\underline{k}) = 0$$

Thus $\underline{b} \times \underline{a}$ is perpendicular to both the vectors $\underline{a}$ and $\underline{b}$.

(iii) $\underline{a} = 3\underline{i} - 2\underline{j} + \underline{k}$, $\underline{b} = \underline{i} + \underline{j}$

Solution:

$$\begin{aligned}\underline{a} \times \underline{b} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 3 & -2 & 1 \\ 1 & 1 & 0 \end{vmatrix} \\ &= \underline{i}(0-1) - \underline{j}(0-1) + \underline{k}(3+2) \\ &= -\underline{i} + \underline{j} + 5\underline{k} \\ \underline{a} \cdot (\underline{a} \times \underline{b}) &= (3\underline{i} - 2\underline{j} + \underline{k}) \cdot (-\underline{i} + \underline{j} + 5\underline{k}) \\ &= 3(-1) + (-2)(1) + 1(5) \\ &= 0 \\ \Rightarrow \underline{a} &\perp \underline{a} \times \underline{b} \\ \underline{b} \cdot (\underline{a} \times \underline{b}) &= (\underline{i} + \underline{j}) \cdot (-\underline{i} + \underline{j} + 5\underline{k}) \\ &= (1)(-1) + (1)(1) \\ &= 0 \\ \Rightarrow \underline{b} &\perp \underline{a} \times \underline{b}\end{aligned}$$

Thus $\underline{a} \times \underline{b}$ is perpendicular to both the vectors $\underline{a}$ and $\underline{b}$.

$$\begin{aligned}\underline{b} \times \underline{a} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 1 & 0 \\ 3 & -2 & 1 \end{vmatrix} \\ &= \underline{i}(1-0) - \underline{j}(1-0) + \underline{k}(-2-3) \\ &= \underline{i} - \underline{j} - 5\underline{k}\end{aligned}$$

$$\begin{aligned}\underline{a} \cdot (\underline{b} \times \underline{a}) &= (3\underline{i} - 2\underline{j} + \underline{k}) \cdot (\underline{i} - \underline{j} - 5\underline{k}) \\ &= 3(1) + (-2)(-1) + (1)(-5) \\ &= 3 + 2 - 5 \\ &= 0\end{aligned}$$

$$\Rightarrow \underline{a} \perp \underline{b} \times \underline{a}$$

$$\begin{aligned}\underline{b} \cdot (\underline{b} \times \underline{a}) &= (\underline{i} + \underline{j}) \cdot (\underline{i} - \underline{j} - 5\underline{k}) \\ &= (1)(1) + (1)(-1) + 0(-5) \\ &= 1 - 1 = 0\end{aligned}$$

$$\Rightarrow \underline{b} \perp \underline{b} \times \underline{a}$$

Thus $\underline{b} \times \underline{a}$ is perpendicular to both the vectors $\underline{a}$ and $\underline{b}$.

(iv) $\underline{a} = -4\underline{i} + \underline{j} - 2\underline{k}$, $\underline{b} = 2\underline{i} + \underline{j} + \underline{k}$

Solution:

$$\begin{aligned}\underline{a} \times \underline{b} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -4 & 1 & -2 \\ 2 & 1 & 1 \end{vmatrix} \\ &= \underline{i}(1+2) - \underline{j}(-4+4) + \underline{k}(-4-2) \\ &= 3\underline{i} - 0\underline{j} - 6\underline{k}\end{aligned}$$

$$\begin{aligned}\underline{a} \cdot (\underline{a} \times \underline{b}) &= (-4\underline{i} + \underline{j} - 2\underline{k}) \cdot (3\underline{i} - 6\underline{k}) \\ &= (-4)(3) + 0 + (-2)(-6) = 0\end{aligned}$$

$$\Rightarrow \underline{a} \perp \underline{a} \times \underline{b}$$

Now

$$\begin{aligned}\underline{b} \cdot (\underline{a} \times \underline{b}) &= (2\underline{i} + \underline{j} + \underline{k}) \cdot (3\underline{i} + 0\underline{j} - 6\underline{k}) \\ &= (2)(3) + (1)(0) + (1)(-6) \\ &= 6 + 0 - 6 \\ &= 0\end{aligned}$$

$$\Rightarrow \underline{b} \perp \underline{a} \times \underline{b}$$

Thus $\underline{a} \times \underline{b}$ is perpendicular to both the vectors $\underline{a}$ and $\underline{b}$.

$$\begin{aligned}\underline{b} \times \underline{a} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & 1 & 1 \\ -4 & 1 & -2 \end{vmatrix} \\ &= \underline{i}(-2-1) - \underline{j}(-4+4) + \underline{k}(2+4) \\ &= -3\underline{i} + 6\underline{k}\end{aligned}$$

Now

$$\begin{aligned}\underline{a} \cdot (\underline{b} \times \underline{a}) &= (-4\underline{i} + \underline{j} - 2\underline{k}) \cdot (-3\underline{i} + 0\underline{j} + 6\underline{k}) \\ &= (-4)(-3) + 1(0) + (-2)(6) \\ &= 0\end{aligned}$$

$$\Rightarrow \underline{a} \perp \underline{b} \times \underline{a}$$

$$\begin{aligned}\text{Also } \underline{b} \cdot (\underline{b} \times \underline{a}) &= (2\underline{i} + \underline{j} + \underline{k}) \cdot (-3\underline{i} + 0\underline{j} + 6\underline{k}) \\ &= (2)(-3) + 1(0) + (1)(6) \\ &= 0\end{aligned}$$

$$\Rightarrow \underline{b} \perp \underline{b} \times \underline{a}$$

Thus $\underline{b} \times \underline{a}$ is perpendicular to both the vectors $\underline{a}$ and $\underline{b}$.

Q.2 Find a unit vector perpendicular to the plane containing $\underline{a}$ and $\underline{b}$. Also find sine of the angle between them.

(i) $\underline{a} = 2\underline{i} - 6\underline{j} - 3\underline{k}$, $\underline{b} = 4\underline{i} + 3\underline{j} - \underline{k}$

Solution:

$$\underline{a} = 2\underline{i} - 6\underline{j} - 3\underline{k}, \underline{b} = 4\underline{i} + 3\underline{j} - \underline{k}$$

$$\begin{aligned}\underline{a} \times \underline{b} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & -6 & -3 \\ 4 & 3 & 1 \end{vmatrix} \\ &= \underline{i}(6+9) - \underline{j}(-2+12) + \underline{k}(6+24) \\ &= 15\underline{i} - 10\underline{j} + 30\underline{k} \\ |\underline{a} \times \underline{b}| &= |15\underline{i} - 10\underline{j} + 30\underline{k}| \\ &= \sqrt{(15)^2 + (-10)^2 + (30)^2} \\ &= \sqrt{225 + 100 + 900} \\ &= \sqrt{1225} \\ &= 35\end{aligned}$$

Let $\hat{n}$ is unit vector

$$\hat{n} = \frac{\underline{a} \times \underline{b}}{|\underline{a} \times \underline{b}|} = \frac{15\underline{i} - 10\underline{j} + 30\underline{k}}{35}$$

$$\hat{n} = \frac{3}{7}\underline{i} - \frac{2}{7}\underline{j} + \frac{6}{7}\underline{k}$$

$$\begin{aligned}\sin \theta &= \frac{|\underline{a} \times \underline{b}|}{|\underline{a}||\underline{b}|} \\ &= \frac{35}{\sqrt{(2)^2 + (-6)^2 + (-3)^2} \sqrt{(4)^2 + (3)^2 + (-1)^2}} \\ &= \frac{35}{\sqrt{4+36+9} \sqrt{16+9+1}} \\ &= \frac{35}{7\sqrt{26}}\end{aligned}$$

$$\boxed{\sin \theta = \frac{5}{\sqrt{26}}}$$

(ii) $\underline{a} = -\underline{i} - \underline{j} - \underline{k}$, $\underline{b} = 2\underline{i} - 3\underline{j} + 4\underline{k}$

Solution:

$$\underline{a} = -\underline{i} - \underline{j} - \underline{k}, \underline{b} = 2\underline{i} - 3\underline{j} + 4\underline{k}$$

$$\begin{aligned}\underline{a} \times \underline{b} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -1 & -1 & -1 \\ 2 & -3 & 4 \end{vmatrix} \\ &= \underline{i}(-4-3) - \underline{j}(-4+2) + \underline{k}(3+2) \\ &= -7\underline{i} + 2\underline{j} + 5\underline{k}\end{aligned}$$

$$\begin{aligned}|\underline{a} \times \underline{b}| &= \sqrt{(-7)^2 + (2)^2 + (5)^2} \\ &= \sqrt{49 + 4 + 25} \\ &= \sqrt{78}\end{aligned}$$

Let $\hat{n}$ is unit vector.

$$\hat{n} = \frac{\underline{a} \times \underline{b}}{|\underline{a} \times \underline{b}|} = \frac{-7\underline{i} + 2\underline{j} + 5\underline{k}}{\sqrt{78}}$$

$$\hat{n} = \frac{-7}{\sqrt{78}}\underline{i} + \frac{2}{\sqrt{78}}\underline{j} + \frac{5}{\sqrt{78}}\underline{k}$$

$$\begin{aligned}\sin \theta &= \frac{|\underline{a} \times \underline{b}|}{|\underline{a}||\underline{b}|} \\ &= \frac{\sqrt{78}}{\sqrt{1+1+1} \sqrt{4+9+16}} \\ &= \frac{\sqrt{78}}{\sqrt{3} \sqrt{29}} \\ &= \frac{\sqrt{3} \sqrt{26}}{\sqrt{3} \sqrt{29}}\end{aligned}$$

$$\boxed{\sin \theta = \frac{\sqrt{26}}{\sqrt{29}}}$$

(iii) $\underline{a} = 2\underline{i} - 2\underline{j} + 4\underline{k}$, $\underline{b} = -\underline{i} + \underline{j} - 2\underline{k}$

Solution:

$$\underline{a} = 2\underline{i} - 2\underline{j} + 4\underline{k}, \underline{b} = -\underline{i} + \underline{j} - 2\underline{k}$$

$$\underline{a} \times \underline{b} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & -2 & 4 \\ -1 & 1 & -2 \end{vmatrix}$$

$$= \underline{i}(4-4) - \underline{j}(-4+4) + \underline{k}(2-2)$$

$$= 0\underline{i} + 0\underline{j} + 0\underline{k}$$

$$|\underline{a} \times \underline{b}| = 0$$

It is not possible to find out required unit vector.

$$\sin \theta = \frac{|\underline{a} \times \underline{b}|}{|\underline{a}||\underline{b}|}$$

$$= \frac{0}{\sqrt{(2)^2 + (-2)^2 + (4)^2} \sqrt{(-1)^2 + (1)^2 + (-2)^2}}$$

$$= \frac{0}{\sqrt{24}\sqrt{6}} = 0$$

$$\boxed{\sin \theta = 0^\circ}$$

(iv) $\underline{a} = \underline{i} + \underline{j}$, $\underline{b} = \underline{i} - \underline{j}$

$$\underline{a} \times \underline{b} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 1 & 0 \\ 1 & -1 & 0 \end{vmatrix}$$

$$= \underline{i}(0-0) - \underline{j}(0-0) + \underline{k}(-1-1)$$

$$= -2\underline{k}$$

$$|\underline{a} \times \underline{b}| = |-2\underline{k}|$$

$$= 2$$

Let $\hat{n}$ is unit vector

$$\hat{n} = \frac{\underline{a} \times \underline{b}}{|\underline{a} \times \underline{b}|} = \frac{-2\underline{k}}{2} = -\underline{k}$$

$$\sin \theta = \frac{|\underline{a} \times \underline{b}|}{|\underline{a}||\underline{b}|}$$

$$= \frac{2}{\sqrt{1+1}\sqrt{1+1}}$$

$$= \frac{2}{\sqrt{2}\sqrt{2}}$$

$$= \frac{2}{2}$$

$$\boxed{\sin \theta = 1}$$

Q.3 Find the area of the triangle, determined by the point P, Q and R.

(i) $P(0,0,0)$, $Q(2,3,2)$, $R(-1,1,4)$

Solution:

$$P(0,0,0), Q(2,3,2), R(-1,1,4)$$

$$\overrightarrow{PQ} = (2\underline{i} + 3\underline{j} + 2\underline{k}) - (0\underline{i} + 0\underline{j} + 0\underline{k})$$

$$= 2\underline{i} + 3\underline{j} + 2\underline{k}$$

$$\overrightarrow{PR} = (-\underline{i} + \underline{j} + 4\underline{k}) - (0\underline{i} + 0\underline{j} + 0\underline{k})$$

$$= -\underline{i} + \underline{j} + 4\underline{k}$$

$$\text{Now } \overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 2 & 3 & 2 \\ -1 & 1 & 4 \end{vmatrix}$$

$$= \underline{i}(12-2) - \underline{j}(8+2) + \underline{k}(2+3)$$

$$= 10\underline{i} - 10\underline{j} + 5\underline{k}$$

$$|\overrightarrow{PQ} \times \overrightarrow{PR}| = \sqrt{(10)^2 + (-10)^2 + (5)^2}$$

$$= \sqrt{100+100+25}$$

$$= \sqrt{225}$$

$$= 15$$

$\therefore$ Area of triangle PQR is

$$\Delta = \frac{1}{2} |\overrightarrow{PQ} \times \overrightarrow{PR}|$$

$$= \frac{1}{2} (15) = \frac{15}{2} \text{ sq. unit}$$

(ii) $P(1,-1,-1)$, $Q(2,0,-1)$, $R(0,2,1)$

Solution:

$$\overrightarrow{PQ} = (2\underline{i} + 0\underline{j} - \underline{k}) - (\underline{i} - \underline{j} - \underline{k})$$

$$= \underline{i} + \underline{j} + 0\underline{k}$$

$$\overrightarrow{PR} = (0\underline{i} + 2\underline{j} + 1\underline{k}) - (\underline{i} - \underline{j} - \underline{k})$$

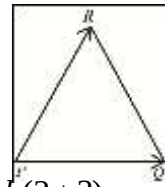
$$= -\underline{i} + 3\underline{j} + 2\underline{k}$$

$$\overrightarrow{PQ} \times \overrightarrow{PR} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 1 & 0 \\ -1 & 3 & 2 \end{vmatrix}$$

$$= \underline{i}(2-0) - \underline{j}(2-0) + \underline{k}(3+1)$$

$$= 2\underline{i} - 2\underline{j} + 4\underline{k}$$

$$|\overrightarrow{PQ} \times \overrightarrow{PR}| = |2\underline{i} - 2\underline{j} + 4\underline{k}|$$



$$\begin{aligned}
 &= \sqrt{(2)^2 + (-2)^2 + (4)^2} \\
 &= \sqrt{4 + 4 + 16} \\
 &= \sqrt{24} \\
 &= 2\sqrt{6}
 \end{aligned}$$

Area of triangle PQR is

$$\begin{aligned}
 \Delta &= \frac{1}{2} |\overrightarrow{PQ} \times \overrightarrow{PR}| \\
 &= \frac{1}{2} (2\sqrt{6}) \\
 &= \sqrt{6} \text{ sq. unit}
 \end{aligned}$$

Q.4 Find the area of parallelogram, whose vertices are:

(i) $A(0,0,0)$, $B(1,2,3)$, $C(2,-1,1)$, $D(3,1,4)$

Solution:

$$\begin{aligned}
 \overrightarrow{AB} &= (\underline{i} + 2\underline{j} + 3\underline{k}) - (0\underline{i} + 0\underline{j} + 0\underline{k}) \\
 &= \underline{i} + 2\underline{j} + 3\underline{k} \\
 \overrightarrow{AC} &= (2\underline{i} - \underline{j} + \underline{k}) - (0\underline{i} + 0\underline{j} + 0\underline{k}) \\
 &= 2\underline{i} - \underline{j} + \underline{k}
 \end{aligned}$$

$$\begin{aligned}
 \overrightarrow{AB} \times \overrightarrow{AC} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & 2 & 3 \\ 2 & -1 & 1 \end{vmatrix} \\
 &= \underline{i}(2+3) - \underline{j}(1-6) + \underline{k}(-1-4) \\
 &= 5\underline{i} + 5\underline{j} - 5\underline{k}
 \end{aligned}$$

Area of parallelogram $ABCD$ is

$$\begin{aligned}
 \Delta &= |\overrightarrow{AB} \times \overrightarrow{AC}| \\
 &= \sqrt{(5)^2 + (5)^2 + (5)^2} \\
 &= \sqrt{25 + 25 + 25} \\
 &= \sqrt{75} \\
 &= 5\sqrt{3} \text{ sq. unit}
 \end{aligned}$$

(ii) $A(1,2,-1)$, $B(4,2,-3)$, $C(6,-5,2)$, $D(9,-5,0)$

Solution:

$$\begin{aligned}
 \overrightarrow{AB} &= (4\underline{i} + 2\underline{j} - 3\underline{k}) - (\underline{i} + 2\underline{j} - \underline{k}) \\
 &= 3\underline{i} + 0\underline{j} - 2\underline{k} \\
 \overrightarrow{AC} &= (6\underline{i} - 5\underline{j} + 2\underline{k}) - (\underline{i} + 2\underline{j} - \underline{k}) \\
 &= 5\underline{i} - 7\underline{j} + 3\underline{k}
 \end{aligned}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 3 & 0 & -2 \\ 5 & -7 & 3 \end{vmatrix}$$

$$\begin{aligned}
 &= \underline{i}(0-14) - \underline{j}(9+10) + \underline{k}(-21-0) \\
 &= -14\underline{i} - 19\underline{j} - 21\underline{k}
 \end{aligned}$$

Area of parallelogram $ABCD$ is

$$\begin{aligned}
 \Delta &= |\overrightarrow{AB} \times \overrightarrow{AC}| \\
 &= \sqrt{(-14)^2 + (-19)^2 + (-21)^2} \\
 &= \sqrt{196 + 361 + 441} \\
 &= \sqrt{998} \text{ sq. unit}
 \end{aligned}$$

(iii) $A(-1,1,1)$, $B(-1,2,2)$,

$C(-3,4,-5)$, $D(-3,5,-4)$

Solution:

$$\begin{aligned}
 \overrightarrow{AB} &= (-\underline{i} + 2\underline{j} + 2\underline{k}) - (-\underline{i} + \underline{j} + \underline{k}) \\
 &= 0\underline{i} + \underline{j} + \underline{k}
 \end{aligned}$$

$$\begin{aligned}
 \overrightarrow{AC} &= (-3\underline{i} + 4\underline{j} - 5\underline{k}) - (-\underline{i} + \underline{j} + \underline{k}) \\
 &= -2\underline{i} + 3\underline{j} - 6\underline{k}
 \end{aligned}$$

$$\overrightarrow{AB} \times \overrightarrow{AC} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 0 & 1 & 1 \\ -2 & 3 & -6 \end{vmatrix}$$

$$\begin{aligned}
 &= \underline{i}(-6-3) - \underline{j}(0+2) + \underline{k}(0+2) \\
 &= -9\underline{i} - 2\underline{j} + 2\underline{k}
 \end{aligned}$$

Area of parallelogram $ABCD$ is

$$\begin{aligned}
 \Delta &= |\overrightarrow{AB} \times \overrightarrow{AC}| \\
 &= \sqrt{(-9)^2 + (-2)^2 + (2)^2} \\
 &= \sqrt{81 + 4 + 4} \\
 &= \sqrt{89} \text{ sq. unit}
 \end{aligned}$$

Q.5 Which vectors, if any, are perpendicular or parallel.

(i) $\underline{u} = 5\underline{i} - \underline{j} + \underline{k}$; $\underline{v} = \underline{j} - 5\underline{k}$,
 $\underline{w} = -15\underline{i} + 3\underline{j} - 3\underline{k}$

Solution:

$$\begin{aligned}\underline{w} &= -15\underline{i} + 3\underline{j} - 3\underline{k} \\ &= -3(5\underline{i} - \underline{j} + \underline{k}) \\ &= -3\underline{u}\end{aligned}$$

$\underline{u}$ and $\underline{w}$ are parallel

(ii) $\underline{u} = \underline{i} + 2\underline{j} - \underline{k}$, $\underline{v} = -\underline{i} + \underline{j} + \underline{k}$,

$$\underline{w} = -\frac{\pi}{2}\underline{i} - \pi\underline{j} + \frac{\pi}{2}\underline{k}$$

Solution:

$$\begin{aligned}\underline{u} \cdot \underline{v} &= (\underline{i} + 2\underline{j} - \underline{k}) \cdot (-\underline{i} + \underline{j} + \underline{k}) \\ &= (1)(-1) + 2(1) + (1)(1) \\ &= -1 + 2 + 1 \\ &= 0\end{aligned}$$

The vectors $\underline{u}$ and $\underline{v}$ are perpendicular

$$\begin{aligned}\underline{v} \cdot \underline{w} &= (-\underline{i} + \underline{j} + \underline{k}) \cdot \left(-\frac{\pi}{2}\underline{i} - \pi\underline{j} + \frac{\pi}{2}\underline{k}\right) \\ &= \frac{\pi}{2} - \pi + \frac{\pi}{2} \\ &= 0\end{aligned}$$

The vectors $\underline{v}$ and $\underline{w}$ are perpendicular

Q.6 Prove that

$$\underline{a} \times (\underline{b} + \underline{c}) + \underline{b} \times (\underline{c} + \underline{a}) + \underline{c} \times (\underline{a} + \underline{b}) = 0$$

Solution:

$$\begin{aligned}\text{L.H.S} &= \underline{a} \times (\underline{b} + \underline{c}) + \underline{b} \times (\underline{c} + \underline{a}) + \underline{c} \times (\underline{a} + \underline{b}) \\ &= \underline{a} \times \underline{b} + \underline{a} \times \underline{c} + \underline{b} \times \underline{c} + \underline{b} \times \underline{a} + \underline{c} \times \underline{a} + \underline{c} \times \underline{b} \\ &= \underline{a} \times \underline{b} + \underline{a} \times \underline{c} + \underline{b} \times \underline{c} + (-\underline{a} \times \underline{b}) + (-\underline{a} \times \underline{c}) + (-\underline{b} \times \underline{c}) \\ &= \underline{a} \times \underline{b} + \underline{a} \times \underline{c} + \underline{b} \times \underline{c} - \underline{a} \times \underline{b} - \underline{a} \times \underline{c} - \underline{b} \times \underline{c} \\ &= 0 \\ &= \text{R.H.S}\end{aligned}$$

Q.7 If $\underline{a} + \underline{b} + \underline{c} = 0$, then prove that

$$\underline{a} \times \underline{b} = \underline{b} \times \underline{c} = \underline{c} \times \underline{a}$$

Solution:

$$\underline{a} + \underline{b} + \underline{c} = 0$$

Taking cross product with $\underline{a}$

$$\underline{a} \times (\underline{a} + \underline{b} + \underline{c}) = 0$$

$$\underline{a} \times \underline{a} + \underline{a} \times \underline{b} + \underline{a} \times \underline{c} = 0$$

$$0 + \underline{a} \times \underline{b} - (\underline{c} \times \underline{a}) = 0$$

$$\underline{a} \times \underline{b} = \underline{c} \times \underline{a} \dots (i)$$

Taking cross product with $\underline{b}$

$$\underline{b} \times (\underline{a} + \underline{b} + \underline{c}) = 0$$

$$\underline{b} \times \underline{a} + \underline{b} \times \underline{b} + \underline{b} \times \underline{c} = 0$$

$$-\underline{a} \times \underline{b} + 0 + \underline{b} \times \underline{c} = 0$$

$$\underline{b} \times \underline{c} = \underline{a} \times \underline{b} \dots (ii)$$

From (i) and (ii)

$$\boxed{\underline{a} \times \underline{b} = \underline{b} \times \underline{c} = \underline{c} \times \underline{a}}$$

Q.8 Prove that $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$

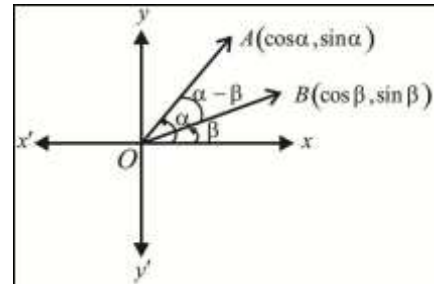
Solution:

Let $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are unit vectors in xy -plane making angle α and β with the positive x -axis respectively.

So that $m\angle BOA = \alpha - \beta$

$$\text{Now } \overrightarrow{OA} = \cos \alpha \underline{i} + \sin \alpha \underline{j}$$

$$\overrightarrow{OB} = \cos \beta \underline{i} + \sin \beta \underline{j}$$



$$\overrightarrow{OB} \times \overrightarrow{OA} = |\overrightarrow{OB}| |\overrightarrow{OA}| \sin(\alpha - \beta) \hat{k} \therefore \overrightarrow{OB} \times \overrightarrow{OA} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ \cos \beta & \sin \beta & 0 \\ \cos \alpha & \sin \alpha & 0 \end{vmatrix}$$

$$\begin{aligned} |\vec{OB}||\vec{OA}|\sin(\alpha - \beta)\hat{k} &= \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(\cos\beta\sin\alpha - \sin\beta\cos\alpha) \quad \therefore |\vec{OA}||\vec{OB}| = 1 \\ \sin(\alpha - \beta)\hat{k} &= (\sin\alpha\cos\beta - \cos\alpha\sin\beta)\hat{k} \\ \sin(\alpha - \beta) &= (\sin\alpha\cos\beta - \cos\alpha\sin\beta) \end{aligned}$$

Q.9 If $\underline{a} \times \underline{b} = 0$ and $\underline{a} \cdot \underline{b} = 0$, what conclusion can be drawn about $\underline{a}$ or $\underline{b}$?

Solution:

If $\underline{a} \times \underline{b} = 0$, then $\underline{a} \parallel \underline{b}$

If $\underline{a} \cdot \underline{b} = 0$ then $\underline{a}$ and $\underline{b}$ are perpendicular, but it is only possible if either $\underline{a} = 0$ or $\underline{b} = 0$ (null vector).

Scalar Triple Product:

For any three vectors $\underline{u}$, $\underline{v}$ and $\underline{w}$, the dot product of one vector with cross product of remaining two vectors is called “Scalar Triple Product” of vectors $\underline{u}$, $\underline{v}$ and $\underline{w}$. It is written as $\underline{u} \cdot (\underline{v} \times \underline{w})$

If $\underline{u} = a_1\underline{i} + b_1\underline{j} + c_1\underline{k}$, $\underline{v} = a_2\underline{i} + b_2\underline{j} + c_2\underline{k}$ and $\underline{w} = a_3\underline{i} + b_3\underline{j} + c_3\underline{k}$ then

$$\begin{aligned} \underline{v} \times \underline{w} &= \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \\ &= \underline{i}(b_2c_3 - c_2b_3) - \underline{j}(a_2c_3 - c_2a_3) + \underline{k}(a_2b_3 - b_2a_3) \\ \underline{u} \cdot (\underline{v} \times \underline{w}) &= (a_1\underline{i} + b_1\underline{j} + c_1\underline{k}) \cdot [\underline{i}(b_2c_3 - c_2b_3) - \underline{j}(a_2c_3 - c_2a_3) + \underline{k}(a_2b_3 - b_2a_3)] \\ &= a_1(b_2c_3 - c_2b_3) - b_1(a_2c_3 - c_2a_3) + c_1(a_2b_3 - b_2a_3) \\ \underline{u} \cdot (\underline{v} \times \underline{w}) &= \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \end{aligned}$$

It is important to note that

$$\underline{u} \cdot (\underline{v} \times \underline{w}) = \underline{v} \cdot (\underline{w} \times \underline{u}) = \underline{w} \cdot (\underline{u} \times \underline{v})$$

The Volume of the Parallelepiped:

The Scalar triple product i.e. $\underline{u} \cdot (\underline{v} \times \underline{w})$ is **volume of a parallelepiped**. Hence it is a scalar.

The Volume of Tetrahedron:

The volume of Tetrahedron = $\frac{1}{6} \underline{u} \cdot (\underline{v} \times \underline{w})$

The properties of Scalar Triple Product:

(i) If $\underline{u}$, $\underline{v}$ and $\underline{w}$ are coplanar then the volume of the parallelepiped is zero that is

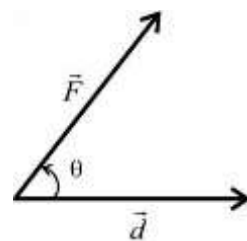
$$(\underline{u} \times \underline{v}) \cdot \underline{w} = 0$$

- (ii) If any two vector of scalar triple product are equal, then its values is zero i.e $[\underline{u} \ \underline{v} \ \underline{v}] = 0$

Work done by a Force:

If a constant Force $\vec{F}$ acts on a body, at an angle θ to the direction of motion, then work done by $\vec{F}$ is define to the product of the component of $\vec{F}$ in the , direction of the displacement and the distance that the body moves.

$$\text{Work done} = \underline{F} \cdot \underline{d} = (\underline{F} \cos \theta) \underline{d} = \underline{F} \underline{d} \cos \theta$$

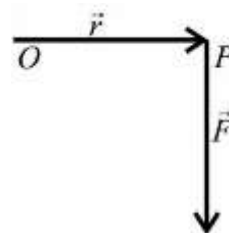


Moment of a Force (Torque):

The turning effect produced by a force is called “Torque” or “Moment” of that Force.

Moment = Perpendicular distance between point of application of force and point of rotation $\times$ Force applied.

$$\begin{aligned} \text{Moment of } \vec{F} \text{ about } O &= \overrightarrow{OP} \times \vec{F} \\ &= \vec{r} \times \vec{F} \end{aligned}$$



EXERCISE 7.5

Q.1 Find the volume of the parallelepiped for which the given vectors are three edges.

- (i) $\underline{u} = 3\underline{i} + 2\underline{k}$, $\underline{v} = \underline{i} + 2\underline{j} + \underline{k}$
 $\underline{w} = -\underline{j} + 4\underline{k}$

Solution:

Volume of parallelepiped is

$$\begin{aligned} \underline{u} \cdot (\underline{v} \times \underline{w}) &= \begin{vmatrix} 3 & 0 & 2 \\ 1 & 2 & 1 \\ 0 & -1 & 4 \end{vmatrix} \\ &= 3(8+1) - 0 + 2(-1-0) \\ &= 27 - 2 \\ &= 25 \text{ cubic unit} \end{aligned}$$

- (ii) $\underline{u} = \underline{i} - 4\underline{j} - \underline{k}$, $\underline{v} = \underline{i} - \underline{j} - 2\underline{k}$,
 $\underline{w} = 2\underline{i} - 3\underline{j} + \underline{k}$

Solution:

Volume of parallelepiped is

$$\begin{aligned} \underline{u} \cdot (\underline{v} \times \underline{w}) &= \begin{vmatrix} 1 & -4 & -1 \\ 1 & -1 & -2 \\ -2 & -3 & 1 \end{vmatrix} \\ &= 1(-1-6) + 4(1+4) - 1(-3+2) \end{aligned}$$

$$= -7 + 20 + 1$$

$$= 14 \text{ cubic unit}$$

- (iii) $\underline{u} = \underline{i} - 2\underline{j} + 3\underline{k}$, $\underline{v} = 2\underline{i} - \underline{j} - \underline{k}$
 $\underline{w} = \underline{j} + \underline{k}$

Solution:

Volume of parallelepiped is

$$\begin{aligned} \underline{u} \cdot (\underline{v} \times \underline{w}) &= \begin{vmatrix} 1 & -2 & 3 \\ 2 & -1 & -1 \\ 0 & 1 & 1 \end{vmatrix} \\ &= 1(-1+1) + 2(2+0) + 3(2+0) \\ &= 0 + 4 + 6 \\ &= 10 \text{ cubic unit} \end{aligned}$$

Q.2 Verify that $\underline{a} \cdot \underline{b} \times \underline{c} = \underline{b} \cdot \underline{c} \times \underline{a} = \underline{c} \cdot \underline{a} \times \underline{b}$ if $\underline{a} = 3\underline{i} - \underline{j} + 5\underline{k}$, $\underline{b} = 4\underline{i} + 3\underline{j} - 2\underline{k}$ and $\underline{c} = 2\underline{i} + 5\underline{j} + \underline{k}$

Solution:

$$\begin{aligned} \underline{a} \cdot \underline{b} \times \underline{c} &= \begin{vmatrix} 3 & -1 & 5 \\ 4 & 3 & -2 \\ 2 & 5 & 1 \end{vmatrix} \\ &= 3(3+10) + 1(4+4) + 5(20-6) \\ &= 39 + 8 + 70 \end{aligned}$$

$$\begin{aligned}
 &= 117 \\
 \underline{b} \cdot \underline{c} \times \underline{a} &= \begin{vmatrix} 4 & 3 & -2 \\ 2 & 5 & 1 \\ 3 & -1 & 5 \end{vmatrix} \\
 &= 4(25+1)-3(10-3)-2(-2-15) \\
 &= 104-21+34 \\
 &= 117
 \end{aligned}$$

$$\begin{aligned}
 \underline{c} \cdot \underline{a} \times \underline{b} &= \begin{vmatrix} 2 & 5 & 1 \\ 3 & -1 & 5 \\ 4 & 3 & -2 \end{vmatrix} \\
 &= 2(2-15)-5(-6-20)+1(9+4) \\
 &= -26+130+13 \\
 &= 117
 \end{aligned}$$

Hence $\underline{a} \cdot \underline{b} \times \underline{c} = \underline{b} \cdot \underline{c} \times \underline{a} = \underline{c} \cdot \underline{a} \times \underline{b}$

Q.3 Prove that the vectors $\underline{i} - 2\underline{j} + 3\underline{k}$, $-2\underline{i} + 3\underline{j} - 4\underline{k}$ and $\underline{i} - 3\underline{j} + 5\underline{k}$ are coplanar.

Solution:

$$\underline{a} = \underline{i} - 2\underline{j} + 3\underline{k}, \underline{b} = -2\underline{i} + 3\underline{j} - 4\underline{k}$$

$$\underline{c} = \underline{i} - 3\underline{j} + 5\underline{k}$$

If vectors $\underline{a}, \underline{b}$ and $\underline{c}$ are coplanar, then we have to show that $\underline{a} \cdot \underline{b} \times \underline{c} = 0$

$$\begin{aligned}
 \underline{a} \cdot \underline{b} \times \underline{c} &= \begin{vmatrix} 1 & -2 & 3 \\ -2 & 3 & -4 \\ 1 & -3 & 5 \end{vmatrix} \\
 &= 1(15-12)+2(-10+4)+3(6-3) \\
 &= 3-12+9=0
 \end{aligned}$$

Thus the given vectors are coplanar.

Q.4 Find the constant α such that the vectors are coplanar.

(i) $\underline{i} - \underline{j} + \underline{k}$, $\underline{i} - 2\underline{j} - \underline{k}$ and $3\underline{i} - \alpha\underline{j} + 5\underline{k}$

Solution:

$$\text{Let } \underline{u} = \underline{i} - \underline{j} + \underline{k}, \underline{v} = \underline{i} - 2\underline{j} - \underline{k} \text{ and}$$

$$\underline{w} = 3\underline{i} - \alpha\underline{j} + 5\underline{k}$$

the vectors $\underline{u}, \underline{v}$ and $\underline{w}$ are coplanar.

If $\underline{u} \cdot \underline{v} \times \underline{w} = 0$

$$\begin{vmatrix} 1 & -1 & 1 \\ 1 & -2 & -3 \\ 3 & -\alpha & 5 \end{vmatrix} = 0$$

$$\begin{aligned}
 &(-10+3\alpha)+1(5+9)+1(-\alpha+6)=0 \\
 &-10-3\alpha+14-\alpha=0 \\
 &-4\alpha+10=0 \\
 &-4\alpha=-10 \Rightarrow \alpha = \frac{5}{2}
 \end{aligned}$$

(ii) $\underline{i} - 2\alpha\underline{j} - \underline{k}$, $\underline{i} - \underline{j} + 2\underline{k}$ and $\alpha\underline{i} - \underline{j} + \underline{k}$

Solution:

$$\text{Let } \underline{u} = \underline{i} - 2\alpha\underline{j} - \underline{k}, \underline{v} = \underline{i} - \underline{j} + 2\underline{k},$$

$$\underline{w} = \alpha\underline{i} - \underline{j} + \underline{k}$$

the vectors $\underline{u}, \underline{v}$ and $\underline{w}$ are coplanar

If $\underline{u} \cdot \underline{v} \times \underline{w} = 0$

$$\begin{vmatrix} 1 & -2\alpha & -1 \\ 1 & -1 & 2 \\ \alpha & -1 & 1 \end{vmatrix} = 0$$

$$1(-1+2)+2\alpha(1-2\alpha)-1(-1+\alpha)=0$$

$$1+2\alpha-4\alpha^2+1-\alpha=0$$

$$-4\alpha^2+\alpha+2=0$$

$$4\alpha^2-\alpha-2=0$$

$$a=4, b=-1, c=-2$$

$$\alpha = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(4)(-2)}}{2(4)}$$

$$= \frac{1 \pm \sqrt{1+32}}{8}$$

$$\alpha = \frac{1 \pm \sqrt{33}}{8}$$

Q.5

(a) Find the value of:

(i) $2\underline{i} \times 2\underline{j} \cdot \underline{k}$

Solution:

$$2\underline{i} \times 2\underline{j} \cdot \underline{k}$$

$$= 2\underline{i} \times 2\underline{j} \cdot \underline{k}$$

$$= 4(\underline{i} \times \underline{j}) \cdot \underline{k}$$

$$= 4(\underline{k} \cdot \underline{k})$$

$$= 4(1)$$

$$= 4$$

(ii) $3\underline{j}.\underline{k} \times \underline{i}$

Solution:

$$\begin{aligned} & 3\underline{j}.\underline{(k \times i)} \\ &= 3\underline{j}.\underline{(j)} \\ &= 3(1) \\ &= 3 \end{aligned}$$

(iii) $\begin{bmatrix} \underline{k} & \underline{i} & \underline{j} \end{bmatrix}$

Solution:

$$\begin{aligned} & \underline{k}.\underline{(i \times j)} \\ &= \underline{k}.\underline{k} \\ &= 1 \end{aligned}$$

(iv) $\begin{bmatrix} \underline{i} & \underline{i} & \underline{k} \end{bmatrix}$

Solution:

$$\begin{aligned} & \underline{i}.\underline{(j \times k)} \\ &= \underline{i}.\underline{(-j)} \\ &= -(\underline{i}.\underline{j}) \\ &= 0 \end{aligned}$$

(b) **Prove that**

$$\begin{aligned} & \underline{u}(\underline{v} \times \underline{w}) + \underline{v}(\underline{w} \times \underline{u}) + \underline{w}(\underline{u} \times \underline{v}) \\ &= 3\underline{u}(\underline{v} \times \underline{w}) \end{aligned}$$

Solution:

$$\begin{aligned} \text{L.H.S} &= \underline{u}.\underline{(v \times w)} + \underline{v}.\underline{(w \times u)} + \underline{w}.\underline{(u \times v)} \\ & \text{we know that} \\ & \underline{u}.\underline{(v \times w)} = \underline{v}.\underline{(w \times u)} = \underline{w}.\underline{(u \times v)} \\ &= \underline{u}.\underline{(v \times w)} + \underline{u}.\underline{(v \times w)} + \underline{u}.\underline{(v \times w)} \\ &= 3\underline{u}.\underline{(v \times w)} \\ &= \text{R.H.S} \end{aligned}$$

Q.6 Find volume of the Tetrahedron with the vertices:

(i) $(0,1,2), (3,2,1), (1,2,1)$ and $(5,5,6)$

Solution:

Let

$$A(0,1,2), B(3,2,1), C(1,2,1), D(5,5,6)$$

$$\begin{aligned} \overrightarrow{AB} &= (3\underline{i} + 2\underline{j} + \underline{k}) - (0\underline{i} + \underline{j} + 2\underline{k}) \\ &= 3\underline{i} + \underline{j} - \underline{k} \end{aligned}$$

$$\begin{aligned} \overrightarrow{AC} &= (\underline{i} + 2\underline{j} + \underline{k}) - (0\underline{i} + \underline{j} + 2\underline{k}) \\ &= \underline{i} + \underline{j} - \underline{k} \end{aligned}$$

$$\begin{aligned} \overrightarrow{AD} &= (5\underline{i} + 5\underline{j} + 6\underline{k}) - (0\underline{i} + \underline{j} + 2\underline{k}) \\ &= 5\underline{i} + 4\underline{j} + 4\underline{k} \end{aligned}$$

Volume of Tetrahedron

$$= \frac{1}{6} [\overrightarrow{AB} \overrightarrow{AC} \overrightarrow{AD}]$$

$$= \frac{1}{6} \begin{vmatrix} 3 & 1 & -1 \\ 1 & 1 & -1 \\ 5 & 4 & 4 \end{vmatrix}$$

$$= \frac{1}{6} [3(4+4) - 1(4+5) - 1(4-5)]$$

$$= \frac{1}{6} [24 - 9 + 1]$$

$$= \frac{1}{6} [16] = \frac{8}{3} \text{ cubic units}$$

(ii) $(2,1,8), (3,2,9), (2,1,4)$ and $(3,3,10)$

Solution:

Let

$$A(2,1,8), B(3,2,9), C(2,1,4), D(3,3,10)$$

$$\begin{aligned} \overrightarrow{AB} &= (3\underline{i} + 2\underline{j} + 9\underline{k}) - (2\underline{i} + \underline{j} + 8\underline{k}) \\ &= \underline{i} + \underline{j} + \underline{k} \end{aligned}$$

$$\begin{aligned} \overrightarrow{AC} &= (2\underline{i} + \underline{j} + 4\underline{k}) - (2\underline{i} + \underline{j} + 8\underline{k}) \\ &= 0\underline{i} + 0\underline{j} - 4\underline{k} \end{aligned}$$

$$\begin{aligned} \overrightarrow{AD} &= (3\underline{i} + 3\underline{j} + 10\underline{k}) - (2\underline{i} + \underline{j} + 8\underline{k}) \\ &= \underline{i} + 2\underline{j} + 2\underline{k} \end{aligned}$$

Volume of Tetrahedron

$$= \frac{1}{6} [\overrightarrow{AB} \overrightarrow{AC} \overrightarrow{AD}]$$

$$= \frac{1}{6} \begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & -4 \\ 2 & 2 & 2 \end{vmatrix}$$

$$= \frac{1}{6} (1(0+8) - 1(0+4) + 1(0-0))$$

$$= \frac{1}{6} (8-4)$$

$$= \frac{1}{6} \times 4$$

$$= \frac{2}{3} \text{ cubic unit}$$

- Q.7** Find the work done, if the point at which the constant force $\underline{F} = 4\underline{i} + 3\underline{j} + 5\underline{k}$ is applied to an object, moves from $P_1(3,1,-2)$ to $P_2(2,4,6)$.

Solution:

$$\begin{aligned}\underline{d} &= \overrightarrow{P_1P_2} \\ &= (2\underline{i} + 4\underline{j} + 6\underline{k}) - (3\underline{i} + \underline{j} - 2\underline{k}) \\ &= -\underline{i} + 3\underline{j} + 8\underline{k}\end{aligned}$$

$$\begin{aligned}\text{Work done} &= \underline{F} \cdot \underline{d} \\ &= (4\underline{i} + 3\underline{j} + 5\underline{k}) \cdot (-\underline{i} + 3\underline{j} + 8\underline{k}) \\ &= 4(-1) + 3(3) + 5(8) \\ &= -4 + 9 + 40 \\ &= 45\end{aligned}$$

- Q.8** A particle, acted by constant forces $4\underline{i} + \underline{j} - 3\underline{k}$ and $3\hat{i} - \hat{j} - \hat{k}$ is displaced from A(1,2,3) to B(5,4,1). Find the work done.

Solution:

$$\begin{aligned}\underline{d} &= \overrightarrow{AB} \\ &= (5\underline{i} + 4\underline{j} + \underline{k}) - (\underline{i} + 2\underline{j} + 3\underline{k}) \\ &= 4\underline{i} + 2\underline{j} - 2\underline{k}\end{aligned}$$

$$\begin{aligned}\underline{F} &= \underline{F}_1 + \underline{F}_2 \\ &= (4\underline{i} + \underline{j} - 3\underline{k}) + (3\underline{i} - \underline{j} - \underline{k}) \\ &= 7\underline{i} + 0\underline{j} - 4\underline{k}\end{aligned}$$

$$\begin{aligned}\text{Work done} &= \underline{F} \cdot \underline{d} \\ &= (7\underline{i} + 0\underline{j} - 4\underline{k}) \cdot (4\underline{i} + 2\underline{j} - 2\underline{k}) \\ &= 7(4) + 0(2) + (-4)(-2) \\ &= 28 + 0 + 8 \\ &= 36\end{aligned}$$

- Q.9** A particle is displaced from the point A(5,-5,-7) to the point B(6,2,2). Under the action of constant forces defined by $10\underline{i} - \underline{j} + 11\underline{k}$, $4\underline{i} + 5\underline{j} + 9\underline{k}$ and $-2\underline{i} + \underline{j} - 9\underline{k}$. Show that the total work done by the forces is 102 units.

Solution:

$$\begin{aligned}\underline{d} &= \overrightarrow{AB} \\ &= (6\underline{i} + 2\underline{j} - 2\underline{k}) - (5\underline{i} - 5\underline{j} - 7\underline{k}) \\ &= \underline{i} + 7\underline{j} + 5\underline{k}\end{aligned}$$

$$\begin{aligned}\underline{F} &= \text{sum of forces} \\ &= (10\underline{i} - \underline{j} + 11\underline{k}) + (4\underline{i} + 5\underline{j} + 9\underline{k}) + (-2\underline{i} + \underline{j} - 9\underline{k}) \\ &= 12\underline{i} + 5\underline{j} + 11\underline{k}\end{aligned}$$

$$\begin{aligned}\text{Work done} &= \underline{F} \cdot \underline{d} \\ &= (12\underline{i} + 5\underline{j} + 11\underline{k}) \cdot (\underline{i} + 7\underline{j} + 5\underline{k}) \\ &= (12)(1) + 5(7) + 11(5) \\ &= 12 + 35 + 55 \\ &= 102 \text{ units}\end{aligned}$$

- Q.10** A force of magnitude 6 units acting parallel to $2\underline{i} - 2\underline{j} + \underline{k}$ displaces the point of application from (1,2,3) to (5,3,7). Find the work done.

Solution:

$$\begin{aligned}\text{Let } \underline{F}_1 &= 2\underline{i} - 2\underline{j} + \underline{k} \\ |\underline{F}_1| &= \sqrt{(2)^2 + (-2)^2 + (1)^2} \\ &= \sqrt{4 + 4 + 1} = \sqrt{9} \\ &= 3\end{aligned}$$

If $\hat{F}_1$ is the unit vector in the direction of $\underline{F}_1$

$$\begin{aligned}\text{Then } \underline{\hat{F}}_1 &= \frac{\underline{F}_1}{|\underline{F}_1|} \\ &= \frac{2\underline{i} - 2\underline{j} + \underline{k}}{3}\end{aligned}$$

The required Force $\underline{F} = 6\underline{\hat{F}}_1$

$$= 6 \left(\frac{2\mathbf{i} - 2\mathbf{j} + \mathbf{k}}{3} \right)$$

$$= 4\mathbf{i} - 4\mathbf{j} + 4\mathbf{k}$$

$$\mathbf{d} = \overrightarrow{AB}$$

$$= (5\mathbf{i} + 3\mathbf{j} + 7\mathbf{k}) - (\mathbf{i} + 2\mathbf{j} + 3\mathbf{k})$$

$$= 4\mathbf{i} + \mathbf{j} + 4\mathbf{k}$$

$$\text{Work done} = \mathbf{F} \cdot \mathbf{d}$$

$$= (4\mathbf{i} - 4\mathbf{j} + 2\mathbf{k}) \cdot (4\mathbf{i} + \mathbf{j} + 4\mathbf{k})$$

$$= 4(4) + (-4)(1) + 2(4)$$

$$= 16 - 4 + 8$$

$$= 20$$

Q.11 A force $\mathbf{F} = 3\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}$ is applied at the point $(1, -1, 2)$. Find the moment of the force about the point $(2, -1, 3)$.

Solution:

$$\text{Let } P(1, -1, 2), Q(2, -1, 3)$$

$$\mathbf{r} = \overrightarrow{QP}$$

$$= (\mathbf{i} - \mathbf{j} + 2\mathbf{k}) - (2\mathbf{i} - \mathbf{j} + 3\mathbf{k})$$

$$= -\mathbf{i} + 0\mathbf{j} - \mathbf{k}$$

Moment of force of $\mathbf{F}$ about Q is

$$\mathbf{r} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 0 & -1 \\ 3 & 2 & -4 \end{vmatrix}$$

$$= \mathbf{i}(0 + 2) - \mathbf{j}(-4 + 3) + \mathbf{k}(-2 - 0)$$

$$= 2\mathbf{i} - 7\mathbf{j} - 2\mathbf{k}$$

Q.12 A force $\mathbf{F} = 4\mathbf{i} - 3\mathbf{k}$, passes through the point $A(2, -2, 5)$. Find the moment of $\mathbf{F}$ about the point $B(1, -3, 1)$.

Solution:

$$\mathbf{r} = \overrightarrow{BA}$$

$$= (2\mathbf{i} - 2\mathbf{j} + 5\mathbf{k}) - (\mathbf{i} - 3\mathbf{j} + \mathbf{k})$$

$$= \mathbf{i} + \mathbf{j} + 4\mathbf{k}$$

Moment of $\mathbf{F}$ about B is

$$\mathbf{r} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 4 \\ 4 & 0 & -3 \end{vmatrix}$$

$$= \mathbf{i}(-3 - 0) - \mathbf{j}(-3 - 16) + \mathbf{k}(0 - 4)$$

$$= -3\mathbf{i} + 19\mathbf{j} - 4\mathbf{k}$$

Q.13 Given a force $\mathbf{F} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ acting at a point $A(1, -2, 1)$. Find the moment of $\mathbf{F}$ about the point $B(2, 0, -2)$.

Solution:

$$\mathbf{r} = \overrightarrow{BA}$$

$$= (\mathbf{i} - 2\mathbf{j} + \mathbf{k}) - (2\mathbf{i} + 0\mathbf{j} + 2\mathbf{k})$$

$$= -\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}$$

$$\mathbf{F} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$$

Moment of force $\mathbf{F}$ about B is

$$\mathbf{r} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & -2 & 3 \\ 2 & 1 & -3 \end{vmatrix}$$

$$= \mathbf{i}(6 - 3) - \mathbf{j}(3 - 6) + \mathbf{k}(1 - 4)$$

$$= 3\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}$$

Q.14 Find the moment about $A(1,1,1)$ of each of the concurrent forces $\underline{i} - 2\underline{j}$, $3\underline{i} + 2\underline{j} - \underline{k}$, $5\underline{j} - 2\underline{k}$, where $P(2,0,1)$ is their point of concurrency.

Solution:

$$\text{Let } \underline{F}_1 = \underline{i} - 2\underline{j} + 0\underline{k}$$

$$\underline{F}_2 = 3\underline{i} + 2\underline{j} - \underline{k}$$

$$\underline{F}_3 = 0\underline{i} + 5\underline{j} - 2\underline{k}$$

$$\underline{F} = \underline{F}_1 + \underline{F}_2 + \underline{F}_3$$

$$= (\underline{i} - 2\underline{j} + 0\underline{k}) + (3\underline{i} + 2\underline{j} - \underline{k}) + (0\underline{i} + 5\underline{j} + 2\underline{k}) \quad \underline{F} = 4\underline{i} + 5\underline{j} + \underline{k}$$

$$\underline{r} = \overrightarrow{AP}$$

$$= (2\underline{i} + 0\underline{j} + \underline{k}) - (\underline{i} + \underline{j} + \underline{k})$$

$$\underline{r} = \underline{i} - \underline{j} + 0\underline{k}$$

Moment of force $\underline{F}$ about A is

$$\underline{r} \times \underline{F} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ 1 & -1 & 0 \\ 4 & 5 & 1 \end{vmatrix}$$

$$= \underline{i}(-1-0) - \underline{j}(1-0) + \underline{k}(5+4)$$

$$= -\underline{i} - \underline{j} + 9\underline{k}$$

Q.15 A force $\underline{F} = 7\underline{i} + 4\underline{j} - 3\underline{k}$ is applied at $P(1,-2,3)$. Find its moment about the point $Q(2,1,1)$.

Solution:

$$\underline{r} = \overrightarrow{QP}$$

$$= (\underline{i} - 2\underline{j} + 3\underline{k}) - (2\underline{i} + \underline{j} + \underline{k})$$

$$\underline{r} = -\underline{i} - 3\underline{j} + 2\underline{k}$$

$$\underline{F} = 7\underline{i} + 4\underline{j} - 3\underline{k}$$

Moment of force $\underline{F}$ about Q is

$$\underline{r} \times \underline{F} = \begin{vmatrix} \underline{i} & \underline{j} & \underline{k} \\ -1 & -3 & 2 \\ 7 & 4 & -3 \end{vmatrix}$$

$$= \underline{i}(9-8) - \underline{j}(3-14) + \underline{k}(-4+21)$$

$$= \underline{i} + 11\underline{j} + 17\underline{k}$$